CS504 QUIZ 3 WITH MOAZ,JUNAID FILES

Final term preparation 2022

ORANGE MONKEY TEAM

1. **Dynamic process model shows the process _____ of the system.**

Structure….confirm

dynamic process model shows the process structure of the system.

2. **Physical view in Krutchen's 4+1 architectural view model captures _____.**

Mapping of the software onto the hardware and reflects is distributed aspect…..confirm

and synchronization aspects of the design, the *physical* view documents the mapping(s) of the software onto the hardware and reflects its distributed aspect, the *development*

3. **A self documenting code is a code that explains itself without the need of comments and extraneous documentation like _____.**

Flowcharts, UML diagrams, process-flow state diagrams…all options correct….confirm

A *self documenting code* is a code that explains itself without the need of comments and extraneous documentation, like flowcharts, UML diagrams, process-flow state diagrams,

4. **Using proper paranthesis normally makes the code.**

All options correct…..confirm

using proper **paranthesis normally m**akes the code.

easy to read

easy to understand

less ambiguous

All of the given options

Software Engineering – 1

(CS504)

Lecture Notes

Delivered by

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ORANGE MONKEY TEAM

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Lecture No. 23

8.6 Architectural Views

Software architecture defines the high level structure of the software by putting together a number of architectural elements in an organized fashion. These elements are chosen to satisfy the functional as well as non-functional requirements of the system. Perry and Wolfe proposed the following formula for software architecture:

Software architecture = {Elements, Forms, Rationale}

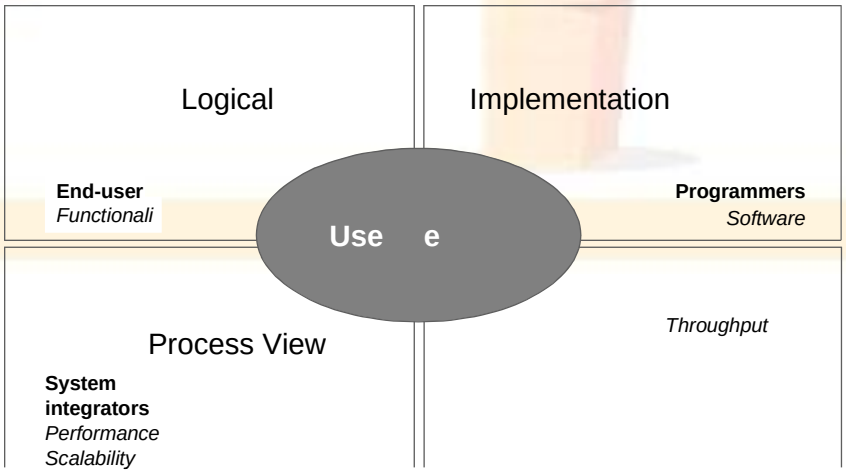
That is, a software architecture is a set of elements which have a certain form. These elements are further divided into three categories. These are: data elements, processing elements, and connecting elements. The data elements contain the information that is used and transformed in the system; the processing elements process the data elements and specify the transformation functions, and connecting elements connect different pieces of architecture together. These can themselves be data or processing elements.

This formula was modified by Boehm as follows:

Software architecture = {Elements, Forms, Rationale/Constraints}

Krutchen proposed that the software architecture model should be composed of multiple views as a software architecture deals with abstraction, with decomposition and composition, with style and esthetics as well as requirements from different stake holders. His architectural model is known as Krutchen's 4+1 architectural view model. As evident, this model proposes the development of 5 main views namely the logical view, the process view, the physical view, the development view, and the use case view. The logical view is the object model of the design, the *process* view captures the concurrency and synchronization aspects of the design, the *physical* view documents the mapping(s) of the software onto the hardware and reflects its distributed aspect, the *development* view describes the static organization of the software in its development environment, and the use case view uses selected use cases or scenarios to validate that the architecture supports the required functionality.

This model is shown in the following diagram.



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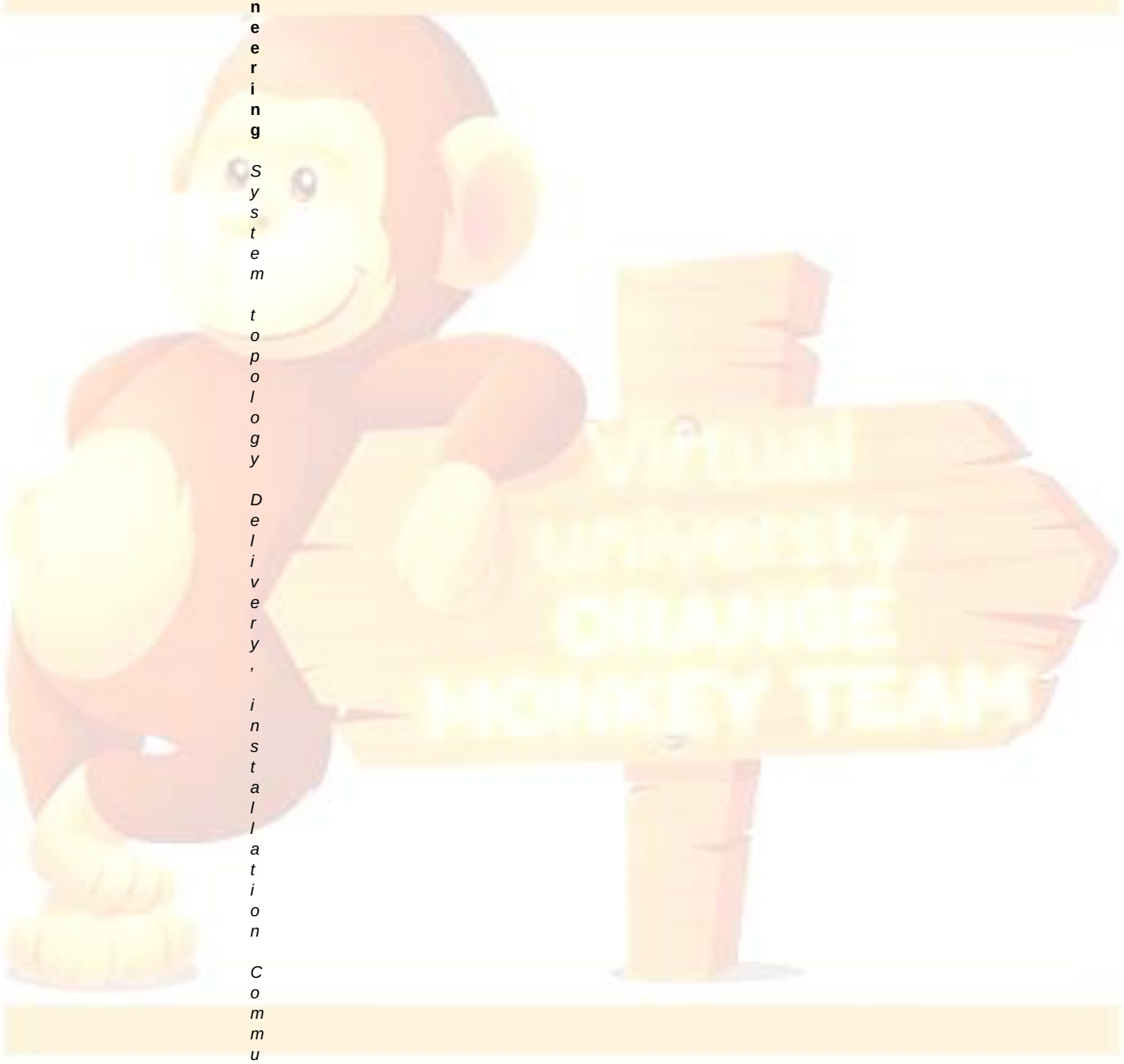
Deployment View

System engineering

System topology

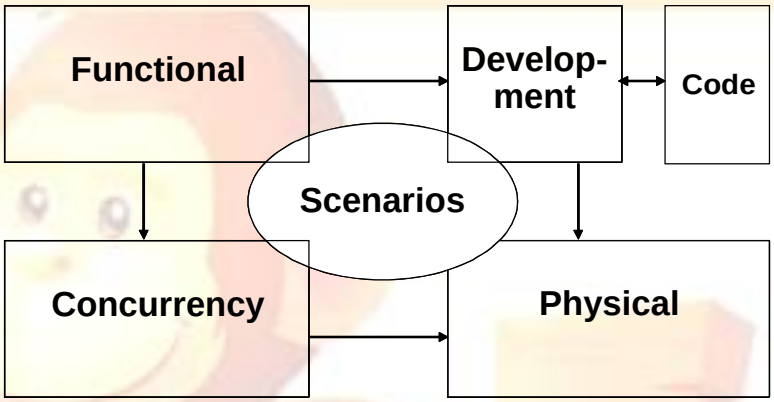
Delivery, installation

Communication



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This model has been slightly modified by Clements et. al. and is shown in the following diagram.



Clement's modified version of Krutchen's 4+1 Architectural View Model

In this model, the architecture is again prepared and analyzed from 5 different perspectives. The 4 main views are Functional View, the Concurrency View, the Physical View, and the Development View. Code view is not present in the original Krutchen model and is basically an extension of the development view.

The Functional View comprises of various different functions provided by the system, key system abstraction, and the domain elements. It connects different dependencies and data flows into a single view. This view can be used by the domain engineers, product-line engineers, as well as the end users of the system. It can be used for understanding the functionality provided by the system, modifiability of the system, reusability, tool support, and allocation of work.

The Development View documents the different files and directories in the system and users of this view include the development and the configuration management staff as well as the project managers. The major uses of this view include maintenance, testing, configuration management, and version control.

The components of the Code View include classes, objects, procedures, functions, subsystems, layers, and modules. It documents the calling as well as the containing hierarchy of different components of the system. Its primary users are the programmers and designers of the system. The primary intent of developing this view is to use it for maintenance and portability.

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Concurrency View intends to document different parallel processes and threads in the system. Its main focus is on event synchronization and parallel data flows. It is used primarily by integrators, performance engineers, and testers. The main purpose of building this view is to identify ways and means to improve performance by highlighting possible opportunities for parallelism.

The Physical View depicts the physical organization of this system and how this system will be physically deployed. This includes different processors, sensors, and storage devices used by the system. This view connects various network elements and communication devices. The primary users of this system include hardware and system engineers. The view is developed with an intention to document and analyze system delivery and installation mechanism. The view is also used in understanding and analyzing issues pertaining to system performance, availability, scalability, and security.

Finally, there are Scenarios. Scenarios are basically use cases which describe the sequences of responsibilities and change cases which are changes to the system. As stated earlier, the first objective of developing a system is to fulfill the functional requirements setout for the system. These functional requirements are written in the form of use cases. Therefore, scenarios are used to understand and validate the system. Their secondary purpose is to communicate the design to the other users of the system. Scenarios tie all the view together into an integrated system. They are also used to understand the dynamic behavior of the system and understand the limits of design.

This is summarized in the following table:

View	Components	Users	Rationale
Functional View	functions, key system abstractions, domain elements	domain engineers, product-line designers, end users	functionality, modifiability, product lines/reusability, tool support, work allocation
Code View	classes, objects, procedures, functions, subsystems, layers, modules	programmers, designers, reusers	modifiability/maintainability, portability, subsetability
Development View	files, directories	managers, programmers, configuration managers	managers, programmers, configuration managers
Physical View	CPUs, sensors, storage	hardware engineers, system engineers	system delivery and installation, performance, availability, scalability, security
Concurrency View	processes, threads	performance engineers, integrators, testers	performance, availability

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What Are Views Used For?

Views are an engineering tool to help achieve desired system qualities. Each view provides an engineering handle on certain quality attributes. In some systems, distinct views collapse into one (e.g., the concurrency and physical views may be the same for small systems.). Views are also used as documentation vehicle for current development and future development. Users of views include both managers and customers. For these reasons views must be *annotated* to support analysis. This annotation can be achieved with the help of scenarios and design *rationale*.

Structures can also be used to document how the current system was developed and how future development should occur. This information is needed by managers and customers.

Although one often thinks about a system's structure in terms of functionality, there are other system properties in addition to functionality (such as physical distribution, process communication, and synchronization) that must be reasoned about at an architectural level. Each structure provides a method for reasoning about some of the relevant quality attributes. The uses structure, for instance, must be engineered (not merely recorded) to build a system that can easily be extended or contracted. The calls structure is engineered to reduce bottlenecks. The module structure is engineered to produce modifiable systems, and so on. Each structure provides a different view into the system and a different leverage point for design to achieve desired qualities in the system.

Hierarchical Views

Every view is potentially hierarchical, e.g. *functional view could contain* sub-functions. Similarly *development view* contains directories which contain files. C*ode view* would have modules and systems that contain sub-modules and sub-systems respectively. C*oncurrency view* contains processes that are further subdivided into threads. Finally, *physical view* clusters contain computers which contain processors.

Architectural views are related to each other in complicated ways. One has to choose the views that are useful to the system being built and to the achievement of qualities that are important for that particular application. The architectural views should be hierarchical (where needed) and should contain enough annotated information to support desired analyses.

Architectural styles

The architectural model of a system may conform to a generic architectural model or style. An awareness of these styles can simplify the problem of defining system architectures. However, most large systems are heterogeneous and do not follow a single architectural style.

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Each style describes a system category that encompasses:

(1) a set of components (e.g., a database, computational modules) that perform a function required by a system,
(2) a set of connectors that enable "communication, coordination and cooperation" among components,
(3) constraints that define how components can be integrated to form the system, and

semantic models that enable a designer to understand the overall properties of a system by analyzing the known properties of its constituent parts.

Lecture No. 24

Architectural models

Like analysis models, many different kinds of architectural models are developed during the architectural design process. Static structural model shows the major system components while a dynamic process model shows the process structure of the system. Interface models are developed to define sub-system interfaces.

Architectural Styles

Architectural design may be based upon a certain pattern of model. These different patterns are also called architectural styles. These styles have different characteristics and attributes and can be useful to solve problems related to a particular situation of requirement. Among the many styles, the most commonly practiced are the following:

- Data-centered architectures
- Client Server Architecture and its variations
- Layered architectures
- Reference Architecture

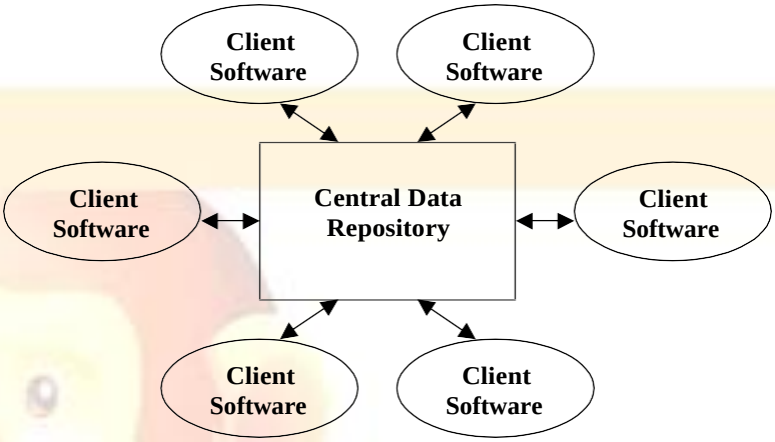
Data-Centered or the repository model

In any system, sub-systems need to exchange information and data. This may be done in two ways:

1. Shared data is held in a central database or repository and may be accessed by all sub-systems
2. Each sub-system maintains its own database and passes data explicitly to other sub-systems

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When large amounts of data are to be shared, the repository model of sharing is most commonly used. This model has been extensively used in the main-frame based



application. The model is depicted in the following diagram:

Repository model characteristics

- Advantages

  Repository model is an efficient way to share large amounts of data. In this case sub-systems need not be concerned with how data is produced Centralised management e.g. backup, security, etc. This model also provides a global view of the system and the sharing model is published as the repository schema.

- Disadvantages

  Repository model suffers from a number of disadvantages. First of all, sub-systems must agree on a repository data model. This inevitably leads to a compromise. The major problem however is that data evolution is difficult and expensive. There is also little scope for implementing specific management policies. It is also difficult to distribute efficiently

Client-server model

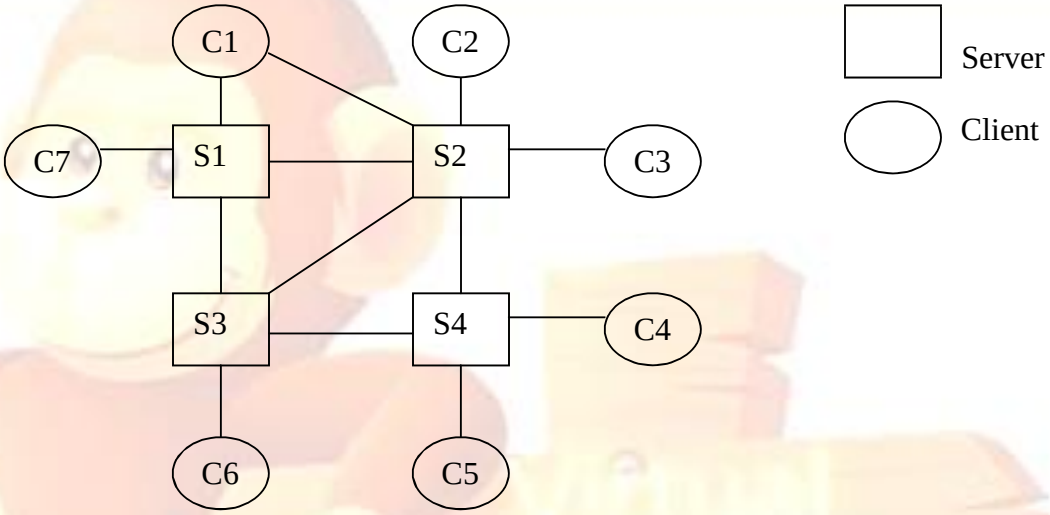
Client server model tries to distribute data and processing. This is a shift from main-frame based applications where both the data management and the processing of data used to be typically carried out by the same main-frame computer. In those applications, the user interface used to be a provided through a "dumb" terminal which did not have much processing power. With the availability of the cheaper but power machines, it was possible to shift some load from the back-end computer to other smaller machines.

The client-server model is a distributed system model which shows how data and processing is distributed across a range of components. In this model, the application is modeled as a set of services that are provided by servers and a set of clients that use these services. The system is organized as a set of stand-alone servers which provide specific services such as printing, data management, etc. and a set of clients which call on these

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services. These clients and servers are connected through a network which allows clients to access servers. Clients and servers are logical processes (not always physical machines). Clients know the servers but the servers do not need to know all the clients and the mapping of processes to processors is not always 1:1.

The following diagram depicts a general client-server organization.



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Client/Server Software Components

A typical client-server architecture based system is composed of a number of different components. These include user interaction/presentation subsystem, application subsystem, database management subsystem, and middleware. The application subsystem implements requirements defined by the application within the context of the operating environment. In this case the application components may reside on either client or server side. Middleware provides the mechanism and protocols to connect clients with the servers.

Representative Client/Server Systems

Following are some of the representative server types in a client-server systems.

- File servers

  In this case, client requests selected records from a file and the server transmits records to client over the network.
- Database servers

  In this case, client sends requests, such as SQL queries, to the database server, the server processes the request and returns the results to the client over the network.
- Transaction servers

  In this configuration, client sends requests that invokes remote procedures on the server side, server executes procedures invoked and returns the results to the client.
- Groupware servers

  Groupware servers provide set of applications that enable communication among clients using text, images, bulletin boards, video, etc.

Client-server characteristics

- Advantages

  The main advantage of the client-server architecture is that it makes effective use of networked systems. Instead of building the system with very expensive large computers and hardware devices, cheaper hardware may be used to gain performance and scalability. In addition, adding new servers or upgrading existing servers becomes easier. Finally, distribution of data is straightforward in this case.

- Disadvantages

  The main disadvantage of this model is that there is no standard way of sharing data, so sub-systems may use different data organisation. Hence data interchange may be inefficient. From a management point of view, each server needs attention and hence there is redundant management in each server. Finally there is no central register of names and services - it may be hard to find out what servers and services are available.

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Representative Client/Server Configurations

The client-server model can have many different configurations. In the following sections, we look at some of these configurations.

Thin Client Model

This model was initially used to migrate legacy systems to client server architectures. In this case the legacy system may act as a server in its own right and the GUI may be implemented on a client. It chief disadvantage is that it places a heavy processing load on both the server and the network.

Fat Client Model

With advent of cheaper and more powerful hardware, people thought of using the processing power of client side machines. So the fat client model came into being. In this model, more processing is delegated to the client as the application processing is locally extended. It is suitable for new client/server systems when the client system capabilities are known in advance. It however is more complex than thin client model with respect to management issues. Since the client machine now also has a significant part of the application resident on it, new versions of each application need to be installed on every client.

Lecture No. 25

Zero Install

As discussed earlier, fat-client architecture posed major challenges in terms of installation and maintenance of the client side of the application, especially when there are large number of client machines. So the idea behind zero install architecture is to develop a system where no installation on the client side is needed. This can only be done when there is no or little processing done at the client side. This is basically a trade-off between using the computing power available at the client machine versus maintenance overhead. This in essence takes us back to an architecture which is similar to thin-client architecture. There is little difference though. In the classical thin-client architecture, the entire processing is carried-out by a single server, in the case of zero-install, the network environment is used to distribute server side processing by adding a number of servers which share processing load. Web-based application where the web-pages are put on a web-server is an example of this type of architecture. In this case, whenever there is a change, the web page is updated accordingly. Now when the user logs in, he gets to use the modified version of the application without any changes or updates at the client side.

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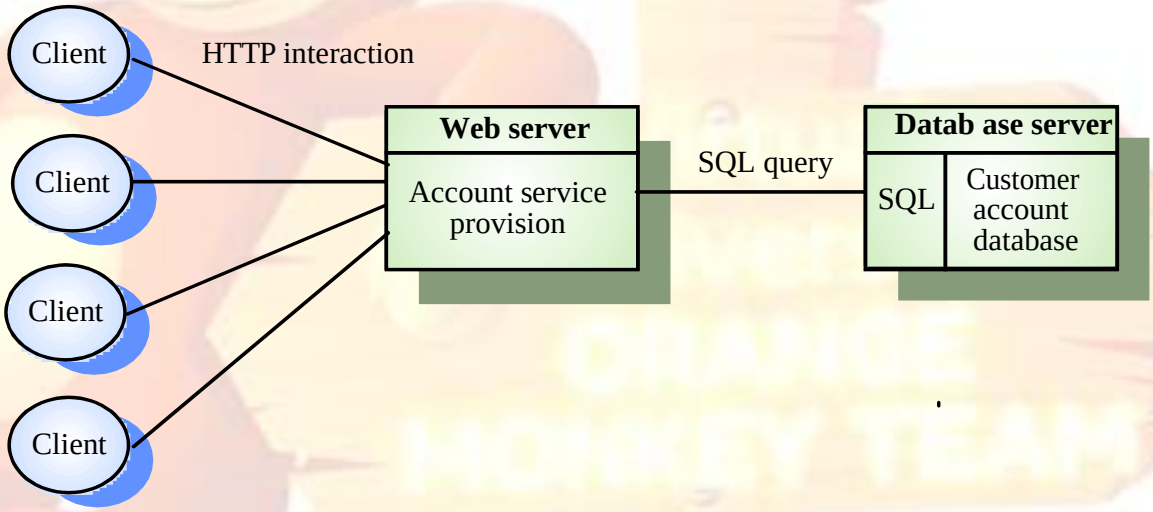
N-Tier architecture

N-tier architecture stems from the struggle to find a middle ground between the fat-client architecture and the thin-client architecture. In this case the idea is to enhance scalability and performance by distributing both the data and the application using multiple server machines. This could involve different types of servers such as application server, web server, and DB server. Three-tier architecture which is explained below is a specialized form of this architecture.

Three-tier Architecture

In this architecture, each application architecture layers (presentation, application, database) may run on separate processors. It therefore allows for better performance than a thin-client approach. It is simpler to manage than fat client approach and highly scalable (as demands increase add more servers).

A typical 3-tier architecture is depicted in the following diagram.



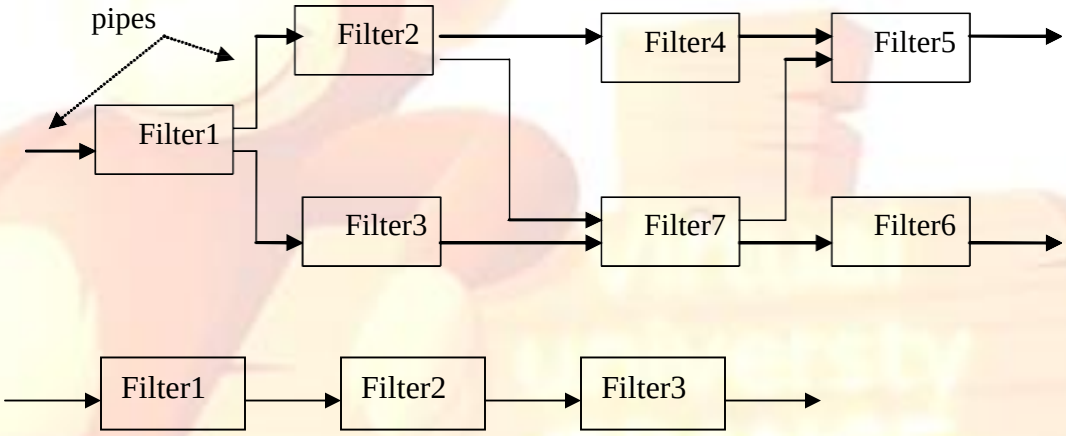
N-tier architecture generalizes the concepts of 3-tier architecture. In this case the system architecture may have more than 3 layers. That is, in n-tier architecture, in order to increase performance, we may distribute the application over different servers by putting different subsystems on different servers.

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Data Flow or Pipes and Filters Architecture

This architecture is very similar to data flow diagrams. This is used when the input data is processed through a series of transformations to yield the desired output. It is also known as pipes and filters architecture where each processing step is called a filter and the connecting link from one process to the other is called the pipe through which the information flows from one process to the other. An important aspect of this model is that each filter works independently of others and does not require knowledge of the working of any of the other filters including its neighbours.

If the dataflow has only a single sequence of processes with no alternative or parallel paths, then it is called batch sequential. These models are depicted in the following diagrams.

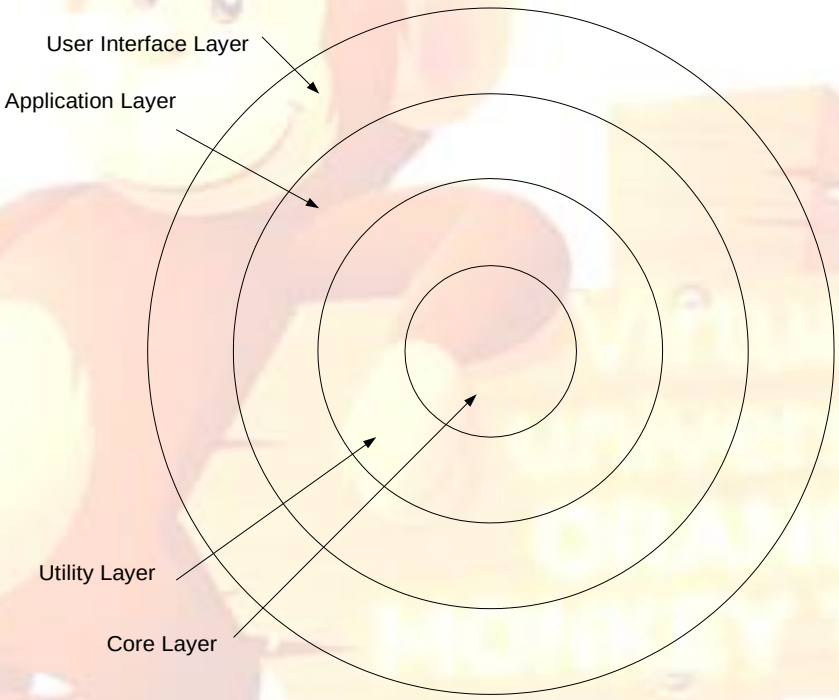


Batch Sequential

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Layered Architecture

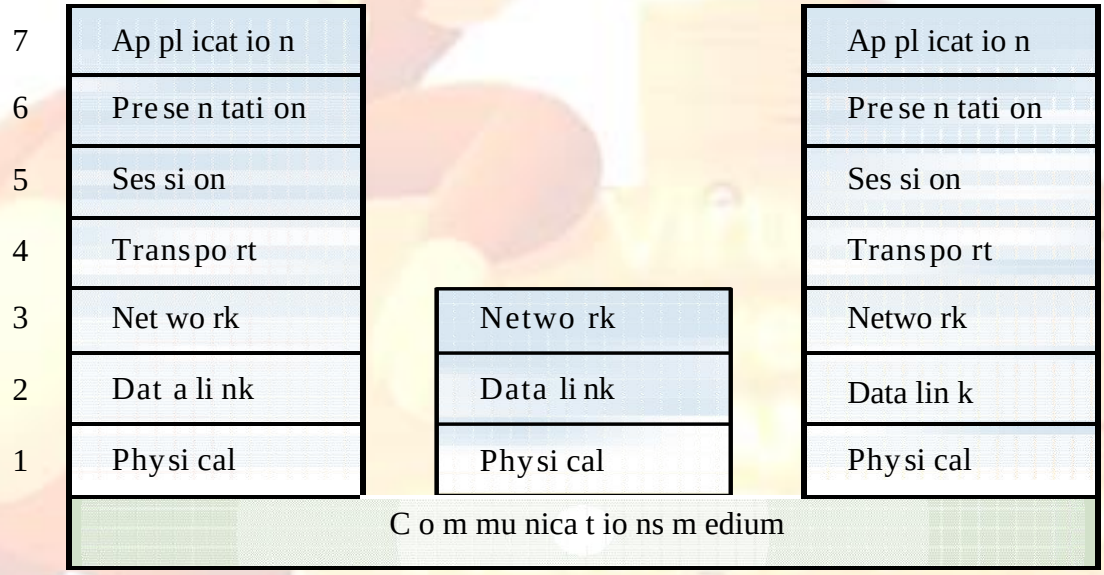
As the name suggests, a layered architecture has many different layers. One typical example of a layered architecture is an operating system where different layers are used to provide services and functionality and the inner layers are closer to the machine hardware than the outer layers. In this way, each layer isolates the outer layer from inner complexities. In order to work properly, the outer layer only needs to know the interface provided by the inner layer. If there are any changes in the inner layer, as long as the interface does not change, the outer layer is not affected. This scheme tremendously portability of the system. The basic layered architecture is depicted in the following diagram.



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Reference architectures

Reference architecture is not a physical architecture. It is only a reference for defining protocols and designing and implementing systems developed by different parties. Reference models are derived from a study of the application domain rather than from existing systems. It may be used as a basis for system implementation or to compare different systems. It acts as a standard against which systems can be evaluated. One very common example of such a reference model is the OSI model which is a layered model for communication systems. The success of this kind model is evident from the success of the Internet where all kinds of heterogeneous systems can talk to each other only because all of them use the same reference architecture.



OSI reference model

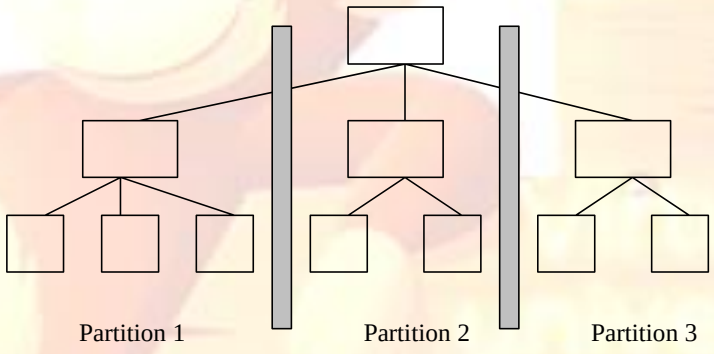
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Partitioning the Architecture

Partitioning of architecture is an important concept. What we basically want to do is distribute the responsibilities to different subsystems so that we get a software system which is easy to maintain. Partitioning results in a system that suffers from fewer side effects. This ultimately means that we get a system that is easier to test and extend and hence is easier to maintain.

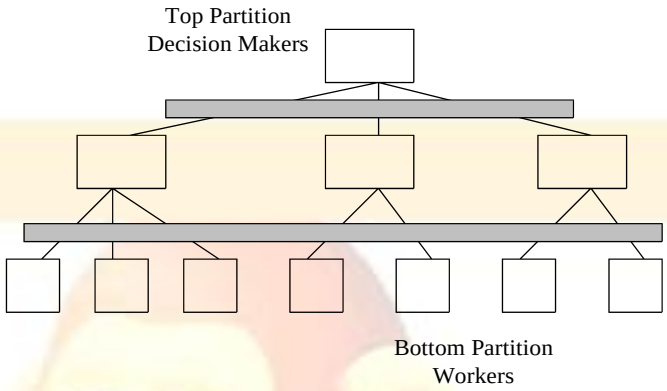
Partitioning of an architecture may be "horizontal" and/or "vertical".

In the horizontal partitioning we define separate branches of the module hierarchy for each major function and control modules are used to coordinate communication between functions. This concept is depicted in the following diagram.



Vertical partitioning divides the application from a decision making perspective. The architecture is partitioned in horizontal layers so that decision making and work are stratified with the decision making modules residing at the top of the hierarchy and worker coming at the bottom. This partitioning is also known as factoring and the general model is depicted in the following diagram.

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Analyzing Architecture design

In a given system, the required characteristics may conflict. Trade-offs seek optimal combinations of properties based on cost/benefit analysis. So the analysis requires an understanding of what is required and the relative priority of the attributes has to be established. The following sequence of steps provides a guideline for performing architectural analysis.

1. Collect scenarios.
2. Elicit requirements, constraints, and environment description.
3. Describe the architectural styles/patterns that have been chosen to address the scenarios and requirements. These views include module view, process view, and data flow view.
4. Evaluate quality attributes by considered each attribute in isolation.
5. Identify the sensitivity of quality attributes to various architectural attributes for a specific architectural style.

Critique candidate architectures (developed in step 3) using the sensitivity analysis conducted in step 5.

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Lecture No. 26

Introduction to Design Patterns

Design Patterns

Christopher Alexander says, "Each pattern describes a problem which occurs over and over again in our environment, and then describes the core of the solution to that problem, in such a way that you can use this solution a million times over, without ever doing it the same way twice." A Pattern Language: Towns/Buildings/Construction, 1977

Even though Alexander was talking about patterns in buildings and towns, what he says is true about object-oriented design patterns. Our solutions are expressed in terms of objects and interfaces instead of walls and doors, but the core of both kinds of patterns is a solution to a problem in a context.

Design Patterns defined

"Description of communicating objects and classes that are customized to solve a general design in a particular context."

Patterns are devices that allow programs to share knowledge about their design. In our daily programming, we encounter many problems that have occurred, and will occur again. The question we must ask our self is how we are going to solve it *this* time. Documenting patterns is one way that you can reuse and possibly share the information that you have learned about how it is best to solve a specific program design problem.

Essay writing is usually done in a fairly well defined form, and so is documenting design patterns. The general form for documenting patterns is to define items such as:

- The motivation or context that this pattern applies to.
- Prerequisites that should be satisfied before deciding to use a pattern.
- A description of the program structure that the pattern will define.
- A list of the participants needed to complete a pattern.
- Consequences of using the pattern...both positive and negative.
- Examples!

Historical perspective of design patterns

The origin of design patterns lies in work done by an architect named *Christopher Alexander* during the late 1970s. He began by writing two books, *A Pattern*

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Language[Alex77] and *A Timeless Way of Building* [Alex79] which, in addition to giving examples, described his rationalle for documenting patterns.

The pattern movement became very quiet until 1987 when patterns appeared again at an OOPSLA conference. Since then, many papers and presentations have appeared, authored by people such as Grady Booch, Richard Helm, and Erich Gamma, and Kent Beck. From then until 1995, many periodicals, featured articles directly or indirectly relating to patterns. In 1995, Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides published Design Patterns: Elements of Reusable Object-Oriented Software [Gamma95], which has been followed by more articles in trade journals.

The concept of design patterns is not new as we can find a number of similar pursuits in the history of program designing and writing. For instance, Standard Template Library (STL) is a library of reusable components provided by C++ compilers. Likewise, we use algorithms in data structures that implement typical operations of manipulating data in data structures. Another, similar effort was from Peter Coad whose patterns are known for object-oriented analysis and design.

Anti-patterns is another concept that corresponds to common mistakes in analysis and design. These are identified in order to prevent potential design and analysis defects from entering into the design. Another, similar concept is object-oriented framework that is a set of cooperative classes that make up reusable design of a system. Framework dictates the architecture of the software and describes the limitations and boundaries of architecture.

With this introduction, we now describe the format that has been adopted in GoF book for describing various design patterns.

GOF Design Pattern Format

The basic template includes ten things as described below

Name

- Works as idiom
- Name has to be meaningful

Problem

- A statement of the problem which describes its intent
- The goals and objectives it wants to reach within the given context

Context

- Preconditions under which the problem and its solutions seem to occur
- Result or consequence
- State or configuration after the pattern has been applied

Forces

- Relevant forces and constraints and their interactions and conflicts.
- motivational scenario for the pattern.

Solution

- Static and dynamic relationships describing how to realize the pattern.
- Instructions on how to construct the work products.

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- Pictures, diagrams, prose which highlight the pattern's structure, participants, and collaborations.

Examples

- One or more sample applications to illustrate
  - a specific context
  - how the pattern is applied
-

Resulting context

- the state or configuration after the pattern has been applied
- consequences (good and bad) of applying the pattern

Rationale

- justification of the steps or rules in the pattern
- how and why it resolves the forces to achieve the desired goals, principles, and philosophies
- how are the forces orchestrated to achieve harmony
- how does the pattern actually work

Related patterns

- the static and dynamic relationships between this pattern and other patterns

Known uses

- to demonstrate that this is a proven solution to a recurring problem

Classifications of deerns

Creational patterns

- How to create and instantiate
- Abstract the instantiation process and make the system independent of its creational process.
- Class creational rules
- Object creational rules
- Abstract factory and factory method
- Abstract the instantiation process
- Make a system independent to its realization
- Class Creational use inheritance to vary the instantiated classes
- Object Creational delegate instantiation to an another object

Structural patterns

- Deals with object's structure
- Class structural patterns concern the aggregation of classes to form the largest classes.
- Object structural patterns concerns the aggregation of objects to form the largest classes
- Class Structural patterns concern the aggregation of classes to form largest structures
- Object Structural pattern concern the aggregation of objects to form largest structures

Behavioral patterns

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- Describe the patterns of communication between classes and objects
- How objects are communicating with each other
- Behavioral class patterns
- Behavioral object patterns
- Use object composition to distribute behavior between classes
- Help in distributing object's intelligence
- Concern with algorithms and assignment of responsibilities between objects
- Describe the patterns of communication between classes or objects
- Behavioral class pattern use inheritance to distribute behavior between classes
- Behavioral object pattern use object composition to distribute behavior between classes

In the following, one pattern from each of the above mentioned categories of design patterns is explained on GoF format.

Lecture No. 27

Observer Pattern

Name

- Observer

Basic intent

- It is intended to define a many to many relationship between objects so that when one object changes state all its dependants are notified and updated automatically.
- Dependence/publish-subscribe mechanism in programming language
    - Smalltalk being the first pure Object Oriented language in which observer pattern was used in implementing its Model View Controller (MVC) pattern. It was a publish-subscribe mechanism in which views (GUIs) were linked with their models (containers of information) through controller objects. Therefore, whenever underlying data changes in the model objects, the controller would notify the view objects to refresh themselves and vice versa.
    - MVC pattern was based on the observer pattern.

Motivation

- It provides a common side effect of partitioning a system into a collection of cooperating classes that are in the need to maintain consistency between related objects.

Description

- This can be used when multiple displays of state are needed.

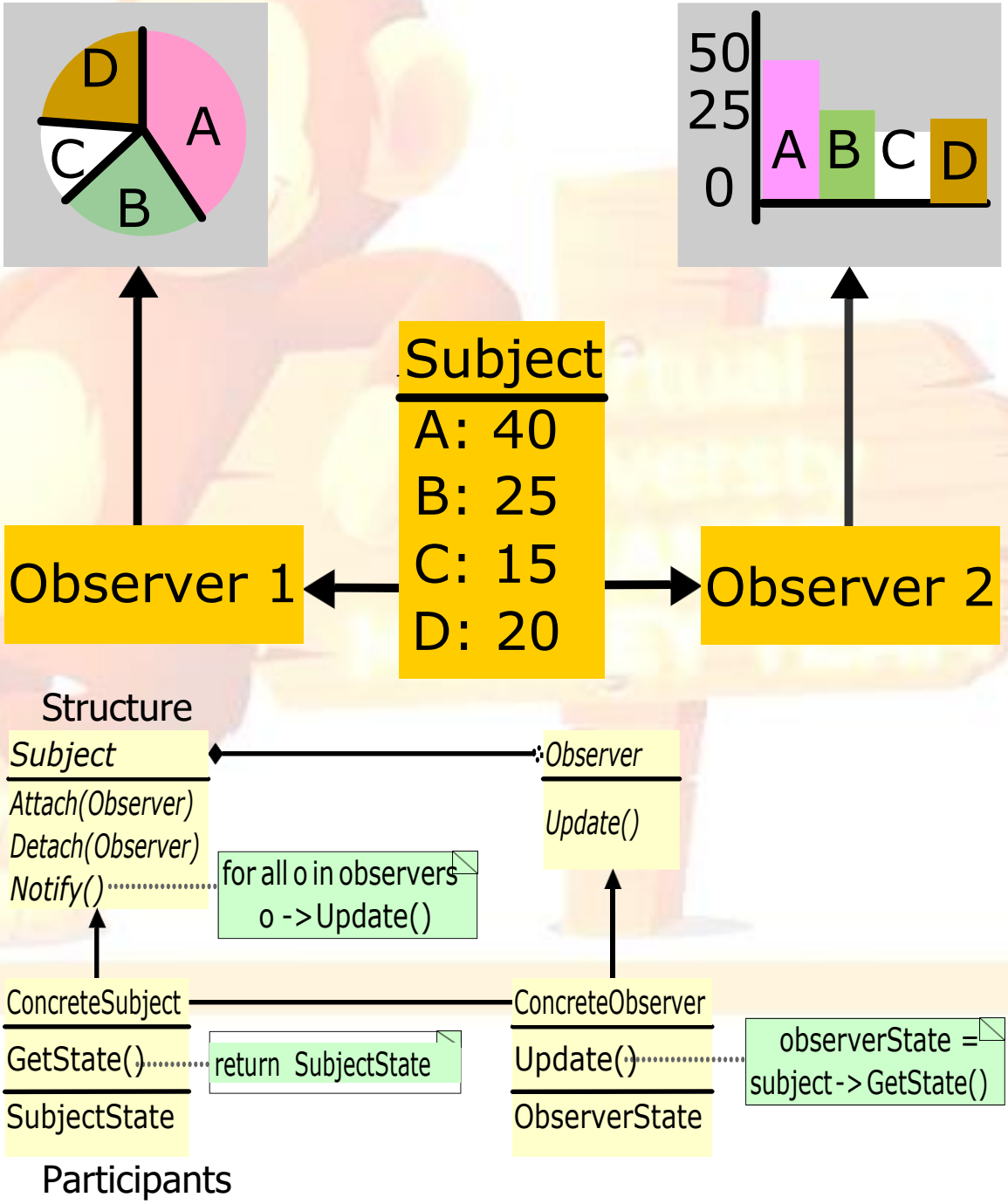
Consequences

- **Optimizations to enhance display performance are impractical.**

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Example implementation of Observer PatternObject Model

Many graphical user interface toolkits separate the presentational aspects of the user interface from the underlying application data. Classes defining application data and presentations can be reused independently. They can work together, too. Both a spreadsheet object and bar chart object can depict information in the same application data object using different presentations. The spreadsheet and the bar chart don't know about each other, thereby letting you reuse only the one you need. But they behave as though they do. When the user changes the information in the spreadsheet, the bar chart reflects the changes immediately, and vice versa.



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Subject

- Knows its observers. Any number of Observer objects may observe a subject.
- Provides an interface for attaching and detaching Observer objects.

Observer

- Defines an updating interface for objects that should be notified of changes in a subject.

ConcreteSubject

- Stores state of interest to concreteObserver objects.
- Sends a notification to its observers when its state changes.

ConcreteObserver

- Maintains a reference to a ConcreteSubject object.
- Stores state state that should stay consistent with the subject's.
- Implements the Observer updating interface to keep its state consistent with the subject's.

Singleton Pattern

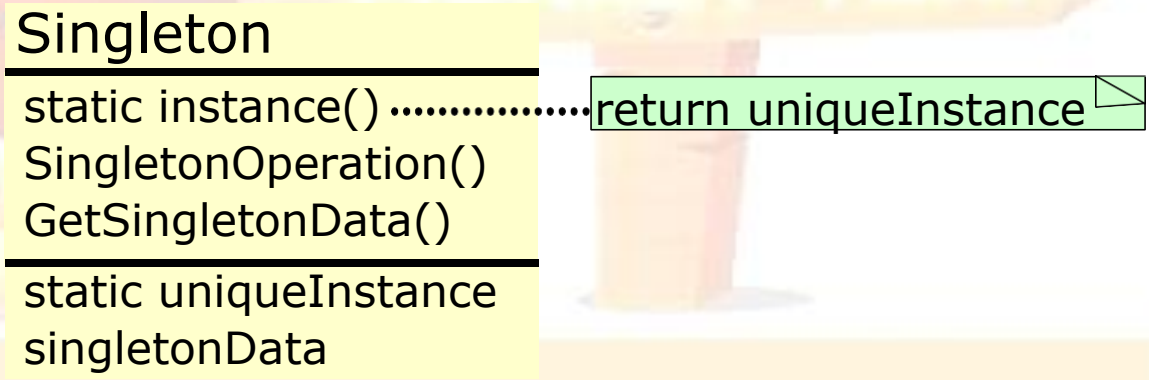
Intent

- It ensures that a class only has one instance and provides a global point of access to it.

Applicability

- Singleton pattern should be used when there must be exactly one instance of a class and it must be accessible to clients from a well-known access point.
- Singleton pattern should be used when controlling the total number of instances that would be created for a particular class.
- Singleton pattern should be used when the sole instance should be extensible by sub classing and clients should be able to use an extended instance without modifying their code.

Structure



Participants

Singleton

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- Defines an instance operation that lets clients access its unique instance. Instance is a class operation (that is, a class method in Smalltalk and a static member function in C++).
- May be responsible for creating its own unique instance.

Singleton Pattern Example

The Singleton class is declared as

```
class Singleton {
public:
    static Singleton* Instance();
protected:
    Singleton();
private:
    static Singleton* _instance;
};
```

The corresponding implementation is

```
Singleton* Singleton::_instance = 0;

Singleton* Singleton::Instance(){
     if (_instance == 0) {
         _instance = new Singleton;
     }
     return _instance;
}
```

Clients access the singleton exclusively through the Instance member function. The variable _instance is initialized to 0, and the static member function Instance returns its value, initializing it with the unique instance if it is 0.

Façade Pattern

Intent

- It provides a unified interface to a set of interfaces in a sub-system.
- Façade defines a higher level interface that makes a subsystem easier to use

Applicability

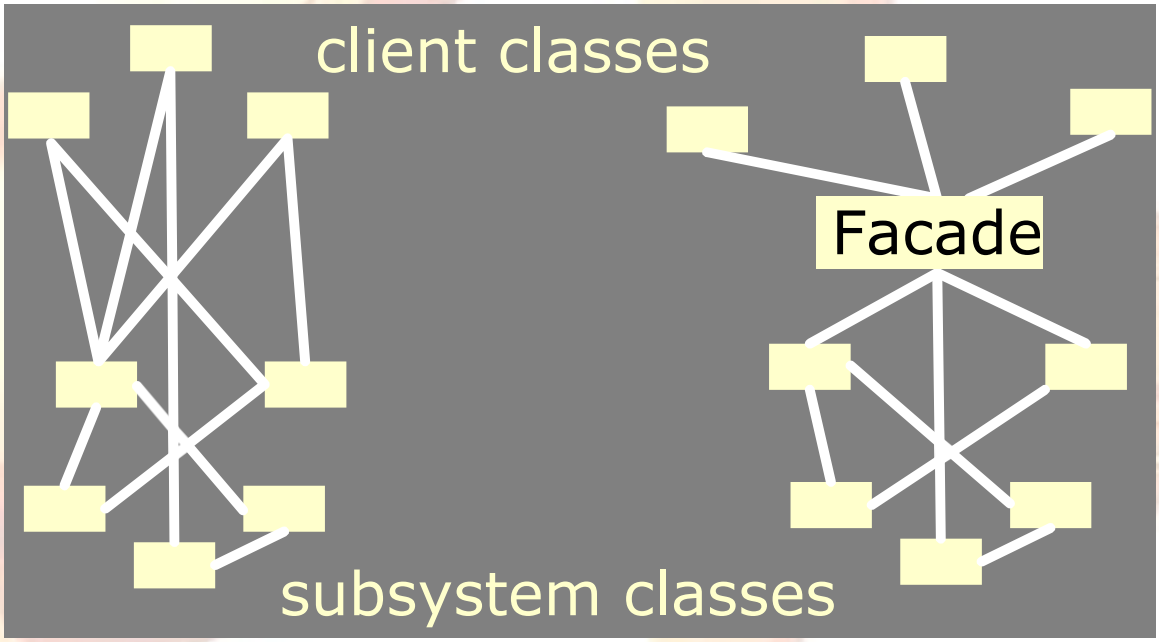
- You would use façade when you want to provide a simple interface to a complex sub-system.
- You would use façade pattern when there are many dependencies between clients and the implementation classes of an abstraction.
- You should introduce a façade to decouple the system from clients and other subsystems.
- You want to layer your subsystem.
- You would use façade when you want to provide a simple interface to a complex sub-system

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- You would use façade pattern when there are many dependencies between clients and the implementation classes of an abstraction
- You should introduce a façade to decouple the system from clients and other subsystems
- You want to layer the subsystem

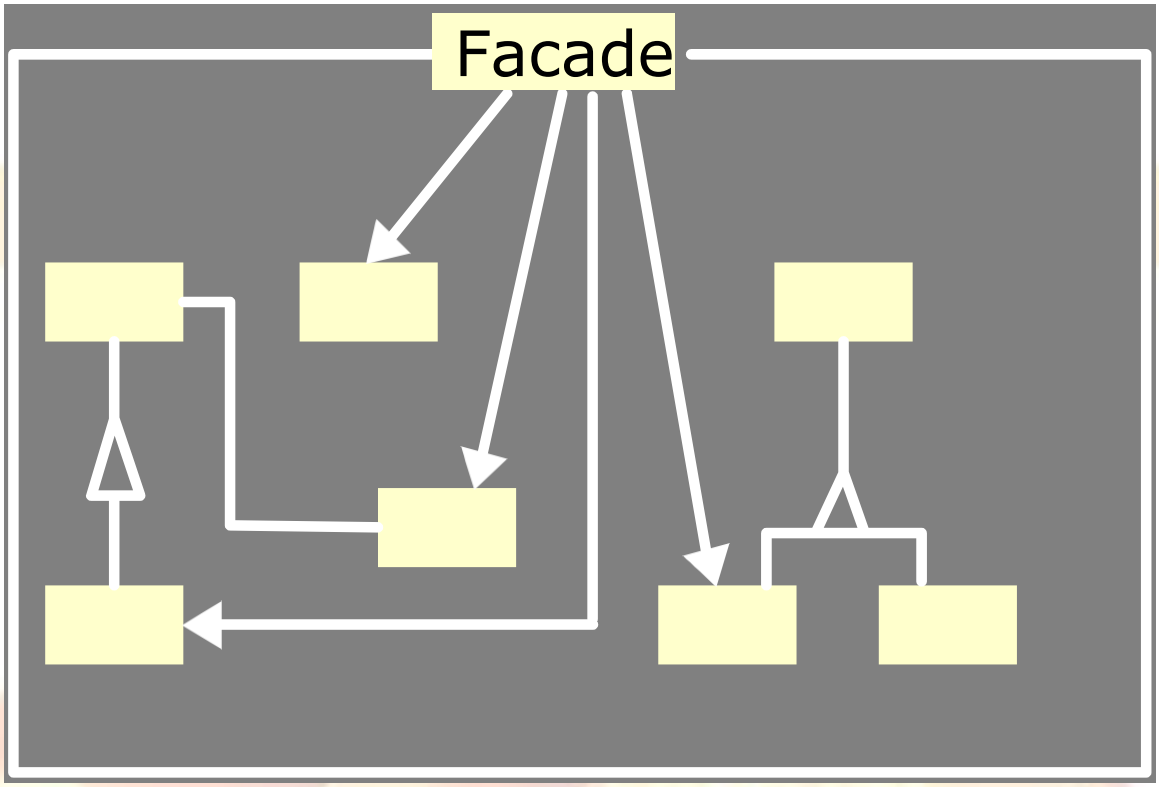
Abstract example of façade

Structuring a system into subsystems helps reduce complexity. A common design goal is to minimize the communication and dependencies between subsystems. One way to achieve this goal is to introduce a façade object that provides a single, simplified interface to the more general facilities of a subsystem.



Structure

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Participants

Façade

- Knows which subsystem classes are responsible for a request.
- Delegates client requests to appropriate subsystem objects.

Subsystem classes

- Implement subsystem functionality.
- Handle work assigned by the Façade object.
- Have no knowledge of the façade, that is, they keep no references to it.

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Lecture No. 28

Good Programming Practices and Guidelines

Maintainable Code

As we have discussed earlier, in most cases, maintainability is the most desirable quality of a software artifact. Code is no exception. Good software ought to have code that is easy to maintain. Fowler says, "*Any fool can write code that computers can understand, good programmers write code that humans can understand."* That is, it is not important to write code that works, it is important to write code that works and is easy to understand so that it can be maintained. The three basic principles that guide maintainability are: simplicity, clarity, and generality or flexibility. The software will be easy to maintain if it is easy to understand and easy to enhance. Simplicity and clarity help in making the code easier to understand while flexibility facilitates easy enhancement to the software.

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Self Documenting Code

From a maintenance perspective, what we need is what is called ***self documenting code***. A *self documenting code* is a code that explains itself without the need of comments and extraneous documentation, like flowcharts, UML diagrams, process-flow state diagrams, etc. That is, the meaning of the code should be evident just by reading the code without having to refer to information present outside this code.

The question is: how can we write code that is self-documenting?

There are a number of attributes that contributes towards making the program self documented. These include, the size of each function, choice of variable and other identifier names, style of writing expressions, structure of programming statements, comments, modularity, and issues relating to performance and portability.

The following discussion tries to elaborate on these points.

Function Size

The size of individual functions plays a significant role in making the program easy or difficult to understand. In general, as the function becomes longer in size, it becomes more difficult to understand. Ideally speaking, a function should not be larger than 20 lines of code and in any case should not exceed one page in length. From where did I get this number 20 and one page? The number 20 is approximately the lines of code that fit on a computer screen and one page of course refers to one printed page. The idea behind these heuristics is that when one is reading a function, one should not need to go back and forth from one screen to the other or from one page to the other and the entire context should be present on one page or on one screen.

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Identifier Names

Identifier names also play a significant role in enhancing the readability of a program. The names should be chosen in order to make them meaningful to the reader. In order to understand the concept, let us look at the following statement.

```
if (x==0) // this is the case when we are allocating a new number
```

In this particular case, the meanings of the condition in the if-statement are not clear and we had to write a comment to explain it. This can be improved if, instead of using x, we use a more meaningful name. Our new code becomes:

```
if (AllocFlag == 0)
```

The situation has improved a little bit but the semantics of the condition are still not very clear as the meaning of 0 is not very clear. Now consider the following statement:

```
If (AllocFlag == NEW_NUMBER)
```

We have improved the quality of the code by replacing the number 0 with a named constant NEW_NUMBER. Now, the semantics are clear and do not need any extra comments, hence this piece of code is self-documenting.

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Coding Style Guide

Consistency plays a very important role in making it self-documenting. A consistently written code is easier to understand and follow. A coding style guide is aimed at improving the coding process and to implement the concept of standardized and relatively uniform code throughout the application or project. As a number of programmers participate in developing a large piece of code, it is important that a consistent style is adopted and used by all. Therefore, each organization should develop a style guide to be adopted by its entire team.

This coding style guide emphasizes on C++ and Java but the concepts are applicable to other languages as well.

Naming Conventions

Hungarian Notation was first discussed by Charles Simonyi of Microsoft. It is a variable naming convention that includes information about the variable in its name (such as data type, whether it is a reference variable or a constant variable, etc). Every company and programmer seems to have their own flavor of Hungarian Notation. The advantage of Hungarian notation is that by just looking at the variable name, one gets all the information needed about that variable.

Bicapitalization or *camel case* (frequently written *CamelCase*) is the practice of writing compound words or phrases where the terms are joined without spaces, and every term is capitalized. The name comes from a supposed resemblance between the bumpy outline of the compound word and the humps of a camel. CamelCase is now the official convention for file names and identifiers in the Java Programming Language.

In our style guide, we will be using a naming convention where Hungarian Notation is mixed with CamelCase.

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General Naming Conventions

General naming conventions for Java and C++

1. Names representing types must be nouns and written in mixed case starting with upper case.

```
Line, FilePrefix
```

2. Variable names must be in mixed case starting with lower case.

```
line, filePrefix
```

This makes variables easy to distinguish from types, and effectively resolves potential naming collision as in the declaration `Line line;`

3. Names representing constants must be all uppercase using underscore to separate words.

```
MAX_ITERATIONS, COLOR_RED
```

In general, the use of such constants should be minimized. In many cases implementing the value as a method is a better choice. This form is both easier to read, and it ensures a uniform interface towards class values.

```
int    getMaxIterations()//    NOT:    MAX_ITERATIONS    =    25
{
                                return                          25;
}
```

4. Names representing methods and functions should be verbs and written in mixed case starting with lower case.

```
getName(), computeTotalWidth()
```

5. Names representing template types in C++ should be a single uppercase letter.

```
template<class                          T>                      ...
template<class C, class D> ...
```

6. Global variables in C++ should always be referred to by using the :: operator.

::mainWindow.open() , ::applicationContext.getName()

7. Private class variables should have _ suffix.

```
class                                                       SomeClass
{
                    private             int                  length_;
                                                                ...
}
```

Apart from its name and its type, the *scope* of a variable is its most important feature. Indicating class scope by using _ makes it easy to distinguish class variables from local scratch variables.

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8. Abbreviations and acronyms should not be uppercase when used as name.

```
exportHtmlSource();          //    NOT:    xportHTMLSource();
openDvdPlayer();       // NOT: openDVDPlayer();
```

Using all uppercase for the base name will give conflicts with the naming conventions given above. A variable of this type would have to be named dVD, hTML etc. which obviously is not very readable.

9. Generic variables should have the same name as their type.

```
void setTopic  (Topic  topic)        // NOT: void setTopic (Topic   value)
                                     // NOT: void setTopic (Topic  aTopic)
                                         // NOT: void setTopic  (Topic  x)

void  connect  (Database  database)  //  NOT:  void  connect  (Database  db)
                                  // NOT: void connect (Database oracleDB)
```

Non-generic variables have a *role*. These variables can often be named by combining role and type:

```
Point                startingPoint,                centerPoint;
Name  loginName;
```

10. All names should be written in English.

```
fileName;     // NOT:    filNavn
```

11. Variables with a large scope should have long names, variables with a small scope can have short names. Scratch variables used for temporary storage or indices are best kept short. A programmer reading such variables should be able to assume that its value is not used outside a few lines of code. Common scratch variables for integers are *i, j, k, m, n* and for characters *c* and *d*.

12. The name of the object is implicit, and should be avoided in a method name.

```
line.getLength();     // NOT:  line.getLineLength();
```

The latter seems natural in the class declaration, but proves superfluous in use.

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Specific Naming Conventions for Java and C++

1. The terms *get/set* must be used where an attribute is accessed directly.

```
employee.getName();
matrix.getElement (2, 4);
employee.setName (name);
matrix.setElement (2, 4, value);
```

2. *is* prefix should be used for boolean variables and methods.

```
isSet, isVisible, isFinished, isFound, isOpen
```

Using the *is* prefix solves a common problem of choosing bad boolean names like *status* or *flag*. *isStatus* or *isFlag* simply doesn't fit, and the programmer is forced to chose more meaningful names.

There are a few alternatives to the *is* prefix that fits better in some situations. These are *has*, *can* and *should* prefixes:

```
boolean hasLicense();
boolean canEvaluate();
boolean shouldAbort = false;
```

3. The term *compute* can be used in methods where something is computed.

```
valueSet.computeAverage();  matrix.computeInverse()
```

Using this term will give the reader the immediate clue that this is a potential time consuming operation, and if used repeatedly, he might consider caching the result.

4. The term *find* can be used in methods where something is looked up.

```
vertex.findNearestVertex();   matrix.findMinElement();
```

This gives the reader the immediate clue that this is a simple look up method with a minimum of computations involved.

5. The term *initialize* can be used where an object or a concept is established.

```
printer.initializeFontSet();
```

6. *List* suffix can be used on names representing a list of objects.

`vertex` (one vertex), `vertexList` (a list of vertices)

Simply using the plural form of the base class name for a list (`matrixElement` (one matrix element), `matrixElements` (list of matrix elements)) should be avoided since the two only differ in a single character and are thereby difficult to distinguish.
A *list* in this context is the compound data type that can be traversed backwards, forwards, etc. (typically a `Vector`). A plain array is simpler. The suffix *Array* can be used to denote an array of objects.

7. *n* prefix should be used for variables representing a number of objects.

```
nPoints, nLines
```


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The notation is taken from mathematics where it is an established convention for indicating a number of objects.

8. *No* suffix should be used for variables representing an entity number.

```
tableNo, employeeNo
```

The notation is taken from mathematics where it is an established convention for indicating an entity number. An elegant alternative is to prefix such variables with an *i*: `iTable, iEmployee`. This effectively makes them *named* iterators.

9. Iterator variables should be called ***i***, ***j***, ***k*** etc.

```
while (Iterator i = pointList.iterator(); i.hasNext(); ) {
  :
}

for (int i = 0; i < nTables; i++) {
  :
}
```

The notation is taken from mathematics where it is an established convention for indicating iterators. Variables named *j*, *k* etc. should be used for nested loops only.

10. Complement names must be used for complement entities.

get/set, add/remove, create/destroy, start/stop, insert/delete, increment/decrement, old/new, begin/end, first/last, up/down, min/max, next/previous, old/new, open/close, show/hide
Reduce complexity by symmetry.

11. Abbreviations in names should be avoided.

```
computeAverage();      // NOT:  compAvg();
```

There are two types of words to consider. First are the common words listed in a language dictionary. These must never be abbreviated. Never write:

`cmd`	instead of	`command`
`cp`	instead of	`copy`
`pt`	instead of	`point`
`comp`	instead of	`compute`
`init`	instead of	`initialize`

etc.

Then there are domain specific phrases that are more naturally known through their acronym or abbreviations. These phrases should be kept abbreviated. Never write:

`HypertextMarkupLanguage`	instead of	`html`
`CentralProcessingUnit`	instead of	`cpu`
`PriceEarningRatio`	instead of	`pe`

etc.

12. Negated boolean variable names must be avoided.

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```
boolean isError;     // NOT:   isNotError
boolean isFound;     // NOT:   isNotFound
```

The problem arise when the logical **NOT** operator is used and double negative arises. It is not immediately apparent what `!isNotError` means.

13. Functions (methods returning an object) should be named after what they return and procedures (*void* methods) after what they do. This increases readability. Makes it clear what the unit should do and especially all the things it is *not* supposed to do. This again makes it easier to keep the code clean of side effects. Naming pointers in C++ specifically should be clear and should represent the pointer type distinctly.

```
Line *line          //NOT Line *pLine; or Line *lineptr; etc
```


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Lecture No. 29

File handling tips for Java and C++

1. C++ header files should have the extension *.h*. Source files can have the extension *.c++* (recommended), *.C*, *.cc* or *.cpp*.

```
MyClass.c++, MyClass.h
```

These are all accepted C++ standards for file extension.

2. Classes should be declared in individual header files with the file name matching the class name. Secondary private classes can be declared as inner classes and reside in the file of the class they belong to. All definitions should reside in source files.

```
class MyClass
{
  public:
    int getValue () {return value_;} // NO!
    ...
  private:
    int value_;
}
```

The header files should declare an interface, the source file should implement it. When looking for an implementation, the programmer should always know that it is found in the source file. The obvious exception to this rule is of course inline functions that must be defined in the header file.

3. Special characters like TAB and page break must be avoided. These characters are bound to cause problem for editors, printers, terminal emulators or debuggers when used in a multi-programmer, multi-platform environment.

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4. The incompleteness of split lines must be made obvious.

```
totalSum = a + b + c +
                   d + e);
function (param1, param2,
                param3);
 setText ("Long line split"
                        "into two parts.");
for (tableNo = 0; tableNo < maxTable;
                      tableNo += tableStep)
```

Split lines occurs when a statement exceed the 80 column limit given above. It is difficult to give rigid rules for how lines should be split, but the examples above should give a general hint.

In general:

- Break after a comma.
- Break after an operator.
- Align the new line with the beginning of the expression on the previous line.



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Include Files and Include Statements for Java and C++

1. Header files must include a construction that prevents multiple inclusion. The convention is an all uppercase construction of the module name, the file name and the h suffix.

```
#ifndef MOD_FILENAME_H
#define MOD_FILENAME_H
 :
#endif
```

The construction is to avoid compilation errors. The construction should appear in the top of the file (before the file header) so file parsing is aborted immediately and compilation time is reduced.

Classes and Interfaces

Class and Interface declarations should be organized in the following manner:

1. Class/Interface documentation.
2. `class` or `interface` statement.
3. Class (**`static`**) variables in the order **`public`**, **`protected`**, package (no access modifier), **`private`**.
4. Instance variables in the order **`public`**, **`protected`**, package (no access modifier), `private`.
5. Constructors.
6. Methods (no specific order).

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Statements in Java and C++

Types

1. Type conversions must always be done explicitly. Never rely on implicit type conversion.

```
floatValue = (float) intValue;      // NOT:    floatValue = intValue;
```

By this, the programmer indicates that he is aware of the different types involved and that the mix is intentional.

2. Types that are local to one file only can be declared inside that file.

3. The parts of a class must be sorted *public*, *protected* and *private*. All sections must be identified explicitly. Not applicable sections should be left out. The ordering is *"most public first"* so people who only wish to use the class can stop reading when they reach the `protected/private` sections.

Variables

1. Variables should be initialized where they are declared and they should be declared in the smallest scope possible.

2. Variables must never have dual meaning. This enhances readability by ensuring all concepts are represented uniquely. Reduce chance of error by side effects.

3. Class variables should never be declared public. The concept of information hiding and encapsulation is violated by public variables. Use private variables and access functions instead. One exception to this rule is when the class is essentially a data structure, with no behavior (equivalent to a C++ `struct`). In this case it is appropriate to make the class' instance variables public.

4. Related variables of the same type can be declared in a common statement. Unrelated variables should not be declared in the same statement.

```
float                 x,                y,                z;
float  revenueJanuary, revenueFebrury, revenueMarch;
```

The common requirement of having declarations on separate lines is not useful in the situations like the ones above. It enhances readability to group variables.

5. Variables should be kept alive for as short a time as possible. Keeping the operations on a variable within a small scope, it is easier to control the effects and side effects of the variable.

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6. Global variables should not be used. Variables should be declared only within scope of their use. Same is recommended for global functions or file scope variables. It is easier to control the effects and side effects of the variables if used in limited scope.

7. Implicit test for *0* should not be used other than for boolean variables and pointers.

```
if (nLines != 0)     // NOT:   if (nLines)
if (value != 0.0)    // NOT:   if (value)
```

It is not necessarily defined by the compiler that ints and floats 0 are implemented as binary 0. Also, by using explicit test the statement give immediate clue of the type being tested. It is common also to suggest that pointers shouldn't test implicit for 0 either, i.e. `if (line == 0)` instead of `if (line)`. The latter is regarded as such a common practice in C/C++ however that it can be used.

Loop structures

1. Only loop control statements must be included in the *for()* construction.

```
sum = 0;                      // NOT:     for (i=0, sum=0; i<100; i++)
for (i=0; i<100; i++)         //              sum += value[i];
  sum += value[i];
```

2. Loop variables should be initialized immediately before the loop.

```
boolean done = false;  // NOT:   boolean  done = false;
while (!done) {        //          :
  :                    //        while (!done) {
}                      //          :
                                  }
```

3. The use of *do while* loops should be avoided. There are two reasons for this. First is that the construct is superflous; Any statement that can be written as a *do while* loop can equally well be written as a *while* lopp or a *for* loop. Complexity is reduced by minimizing the number of constructs being used. The other reason is of readability. A loop with the conditional part at the end is more difficult to read than one with the conditional at the top.

4. The use of *break* and *continue* in loops should be avoided. These statements should only be used if they prove to give higher readability than their structured counterparts. In general *break* should only be used in *case* statements and *continue* should be avoided alltogether.

5. The form *for (;;)* should be used for empty loops.

```
for (;;) {             // NOT:   while (true) {
  :                    //          :
}                      //        }
```


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This form is better than the functionally equivalent *while (true)* since this implies a test against *true*, which is neither necessary nor meaningful. The form while(true) should be used for infinite loops.

Conditionals

1. Complex conditional expressions must be avoided. Introduce temporary boolean variables instead.

```
if ((elementNo < 0) || (elementNo > maxElement)||
    elementNo == lastElement) {
  :
}
```

should be replaced by:

```
boolean isFinished      = (elementNo < 0) || (elementNo > maxElement);
boolean isRepeatedEntry = elementNo == lastElement;
if (isFinished || isRepeatedEntry) {
  :
}
```

2. The nominal case should be put in the *if*-part and the exception in the *else*-part of an if statement.

```
boolean isError = readFile (fileName);
if (!isError) {
  :
}
else {
  :
}
```

3. The conditional should be put on a separate line.

```
if (isDone)               // NOT:  if (isDone) doCleanup();
  doCleanup();
```

4. Executable statements in conditionals must be avoided.

```
file = openFile (fileName, "w");   // NOT:    if ((file = openFile
(fileName, "w")) != null) {
if (file != null) {                //           :
  :                                //         }
}
```


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Miscellaneous

1. The use of magic numbers in the code should be avoided. Numbers other than *0* and *1* should be considered declared as named constants instead.

2. Floating point constants should always be written with decimal point and at least one decimal.

```
double total = 0.0;    // NOT: double total = 0;
double speed = 3.0e8; // NOT: double speed = 3e8;

double sum;
:
sum = (a + b) * 10.0;
```

This emphasizes the different nature of integer and floating point numbers even if their values might happen to be the same in a specific case. Also, as in the last example above, it emphasize the type of the assigned variable (`sum`) at a point in the code where this might not be evident.

3. Floating point constants should always be written with a digit before the decimal point.

```
double total = 0.5;    // NOT: double total = .5;
```

The number and expression system in Java is borrowed from mathematics and one should adhere to mathematical conventions for syntax wherever possible. Also, 0.5 is a lot more readable than .5; There is no way it can be mixed with the integer 5.

4. Functions in C++ must always have the return value explicitly listed.

```
int getValue()      // NOT: getValue()
{
  :
}
```

If not explicitly listed, C++ implies `int` return value for functions.

5. `goto` in C++ should not be used. Goto statements violates the idea of structured code. Only in some very few cases (for instance breaking out of deeply nested structures) should goto be considered, and only if the alternative structured counterpart is proven to be less readable.

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Lecture No. 30

Layout and Comments in Java and C++

Comments

The problem with comments is that they lie. Comments are not syntax checked, there is nothing forcing them to be accurate. And so, as the code undergoes change during schedule crunches, the comments become less and less accurate.

As Fowler puts it, comments should not be used as deodorants. Tricky code should not be commented but rewritten. In general, the use of comments should be minimized by making the code self-documenting by appropriate name choices and an explicit logical structure.

If, however, there is a need to write comments for whatever reason, the following guidelines should be observed.

1. All comments should be written in English. In an international environment English is the preferred language.

2. Use // for all comments, including multi-line comments.

```
// Comment spanning
// more than one line
```

Since multilevel commenting is not supported in C++ and Java, using // comments ensure that it is always possible to comment out entire sections of a file using /* */ for debugging purposes etc.

3. Comments should be indented relative to their position in the code.

```
while (true) {          // NOT:    while (true) {
  // Do something       //         // Do something
  something();          //             something();
}                       //         }
```


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Expressions and Statements

Layout

1. Basic indentation should be 2.

```
for (i = 0; i < nElements; i++)
  a[i] = 0;
```

Indentation of 1 is to small to emphasize the logical layout of the code. Indentation larger than 4 makes deeply nested code difficult to read and increase the chance that the lines must be split. Choosing between indentation of 2, 3 and 4, 2 and 4 are the more common, and 2 chosen to reduce the chance of splitting code lines.

Natural form for expression

Expression should be written as if they are written as comments or spoken out aloud. Conditional expression with negation are always difficult to understand. As an example consider the following code:

```
if (! (block < activeBlock) || !(blockId >= unblocks))
```

The logic becomes much easier to follow if the code is written in the natural form as shown below:

```
if ((block >= activeBlock) || (blockId < unblocks))
```

Parenthesize to remove ambiguity

Parentheses should always be used as they reduce complexity and clarify things by specifying grouping. It is especially important to use parentheses when different unrelated operators are used in the same expression as the precedence rules are often assumed by the programmers, resulting in logical errors that are very difficult to spot. As an example consider the following statement:

```
if (x & MASK == BITS)
```

This causes problems because == operator has higher precedence than & operator. Hence, MASK and BITS are first compared for equality and then the result, which is 0 or 1, is andded with x. This kind of error will be extremely hard to catch. If, however, parentheses are used, there will be no ambiguity as shown below.

```
if ((x & MASK) == BITS)
```


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Following is another example of the use of parentheses which makes the code easier to understand and hence easier to maintain.

```
leapYear = year % 4 == 0 && year % 100 != 0 || year % 400 == 0 ;
```

In this case parentheses have not been used and therefore the definition of a leap year is not very clear for the code. The code becomes self explanatory with the help of proper use of parentheses as shown below:

```
leapYear = ((year % 4 == 0) && (year % 100 != 0)) ||
                    (year % 400 == 0);
```

Breakup complex expressions

Complex expressions should be broken down into multiple statements. An expression is considered to be complex if it uses many operators in a single statement. As an example consider the following statement:

```
*x += (*xp=(2*k < (n-m) ? c[k+1] : d[k--]));
```

This statement liberally uses a number of operators and hence is very difficult to follow and understand. If it is broken down into simple set of statements, the logic becomes easier to follow as shown below:

```
if (2*k < n-m)
        *xp = c[k+1];
else
        *xp = d[k--];
*x = *x + *xp;
```

Shortcuts and cryptic code

Sometimes the programmers, in their creative excitement, try to write very concise code by using shortcuts and playing certain kinds of tricks. This results in a code which is cryptic in nature and hence is difficult to follow. Maintenance of such code therefore becomes a nightmare. Following are some examples of such code.

1. Let us start with a very simple shortcut, often used by programmers. Assume that we have the following statement.

```
x *= a;
```

For some reason the code was later modified to:

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```
x *= a + b;
```

This seemingly harmless change is actually a little cryptic and causes confusion. The problem lies with the semantics of this statement. Does it mean x = x*a+b or x = x*(a+b)? The second one is the right answer but is not obvious from the syntax and hence causes problems.

2. Let us now look at a more complex example. What is the following code doing?

```
subkey = subkey >> (bitoff – (bitoff >> 3) << 3));
```

As can be seen, it is pretty hard to understand and therefore difficult to debug in case there is any problem. What this code is actually doing is masking bitoff with octal 7 and then use the result to shift subkey those many time. This can be written as follows:

```
subkey = subkey >> (bitoff & 0x7);
```

It is quite evident that the second piece of code is much follow to read than the first one.

3. The following piece of code is taken from a commercial software:

```
a = a >> 2;
```

It is easy to see that a is shifted right two times. However, the real semantics of this code are hidden – the real intent here is to divide a by 4. No doubt that the code above achieves this objective but it is hard for the reader to understand the intent as to why a is being shifted right twice. It would have been much better had the code been written as follows:

```
a = a/4;
```

4. A piece of code similar to the following can be found in many data structures books and is part of circular implementation of queues using arrays.

```
bool Queue::add(int n)
{
        int k = (rear+1) % MAX_SIZE;
        if (front == k)
                return false;
        else {
                rear = k;
                queue[rear] = n;
                return true;
        }
}
```


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```
bool Queue::isFull()
{
        if (front == (rear+1 % MAX_SIZE))
                return true;
        else
                return false;
}
```

This code uses the % operator to set the rear pointer to 0 once it has reached MAX_SIZE. This is not obvious immediately. Similarly, the check for queue full is also not trivial.

In an experiment, the students were asked to implement double-ended queue in which pointer could move in both directions. Almost everyone made the mistake of writing something like rear = (rear-1) % MAX_SIZE. This is because the semantics of % operation are not obvious.

It is always much better to state the logic explicitly. Also, counting is also much easier to understand and code as compared to some tricky comparisons (e.g. check for isFull in this case). Application of both these principles resulted in the following code. It is easy to see that this code is easier to understand. It is interesting to note that when another group of students were asked to do implement double-ended queue after showing them this code, almost everyone did it without any problems.

```
bool Queue::add()
{
        if (! isFull() ) {
                rear++;
                if (rear == MAX_SIZE) rear = 0;
                QueueArray[rear] = n;
                size++;
                return true;
        }
        else return false;
}

bool Queue::isFull(int n)
{
        if (size == MAX_SIZE)
                return true;
        else
                return false;
}
```


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Lecture No. 31

Coding Style Guidelines (Continued)

Switch Statement

In the switch statement, cases should always end with a break. A tricky sequence of fall-through code like the one below causes more trouble than being helpful.

```
switch(c) {
case '-' : sign = -1;
case '+' : c = getchar();
case '.' : break;
default : if (! isdigit(c))
                    return 0;
}
```

This code is cryptic and difficult to read. It is much better to explicitly write what is happening, even at the cost of duplication.

```
switch(c) {
case '-': sign = -1;
                    c = getchar();
                    break;
case '+':    c = getchar();
                    break;
case '.': break;
default:     if (! isdigit(c))
                        return 0;
                    break;
}
```

It would even be better if such code is written using the if statement as shown below.

```
if (c == '-') {
        sign = -1;
        c = getchar();
}
else if (c == '+') {
        c = getchar();
}
else if (c != '.' && !isdigit(c)) {
        return 0;
}
```


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Magic Numbers

Consider the following code segment:

```
fac = lim / 20;
if (fac < 1)
        fac = 1;
for (i =0, col = 0; i < 27; i++, j++) {
        col += 3;
        k = 21 – (let[i] /fac);
        star = (let[i] == 0) ? ‘ ’ : ‘*’;
        for (j = k; j < 22; j++)
                draw(j, col, star);
}
draw(23, 1, ‘ ’);
for (i=‘A’; i <= ‘Z’; i++)
        cout << i;
```

Can you tell by reading the code what is meant by the numbers 20, 27, 3, 21, 22, and 23. These are constant that mean something but they do not give any indication of their importance or derivation, making the program hard to understand and modify. To a reader they work like magic and hence are called magic numbers. Any number (even 0 or 1) used in the code is a magic number. It should rather have a name of its own that can be used in the program instead of the number.

The difference would be evident if we look at the code segment below that achieves the same purpose as the code above.

```
enum {
        MINROW = 1,
        MINCOL = 1,
        MAXROW = 24,
        MAXCOL = 80,
        LABELROW = 1,
        NLET = 26,
        HEIGHT = MAXROW –4,
        WIDTH = (MAXCOL-1) / NLET
};

fac = (lim+HEIGHT-1) /HEIGHT;
if (fac < 1)
        fac = 1;
for (i =0; i < NLET; i++) {
        if (let[i] == 0)
                continue;
        for (j = HEIGHT – let[i] / fac; j < HEIGHT; j++)
                draw(j-1 + LABELROW,
                        (i+1)*WIDTH, ‘*’);
}

draw(MAXROW-1, MINCOL+1, ‘ ’);
```


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```
for (i='A'; i <= 'Z'; i++)
        cout << i;
```

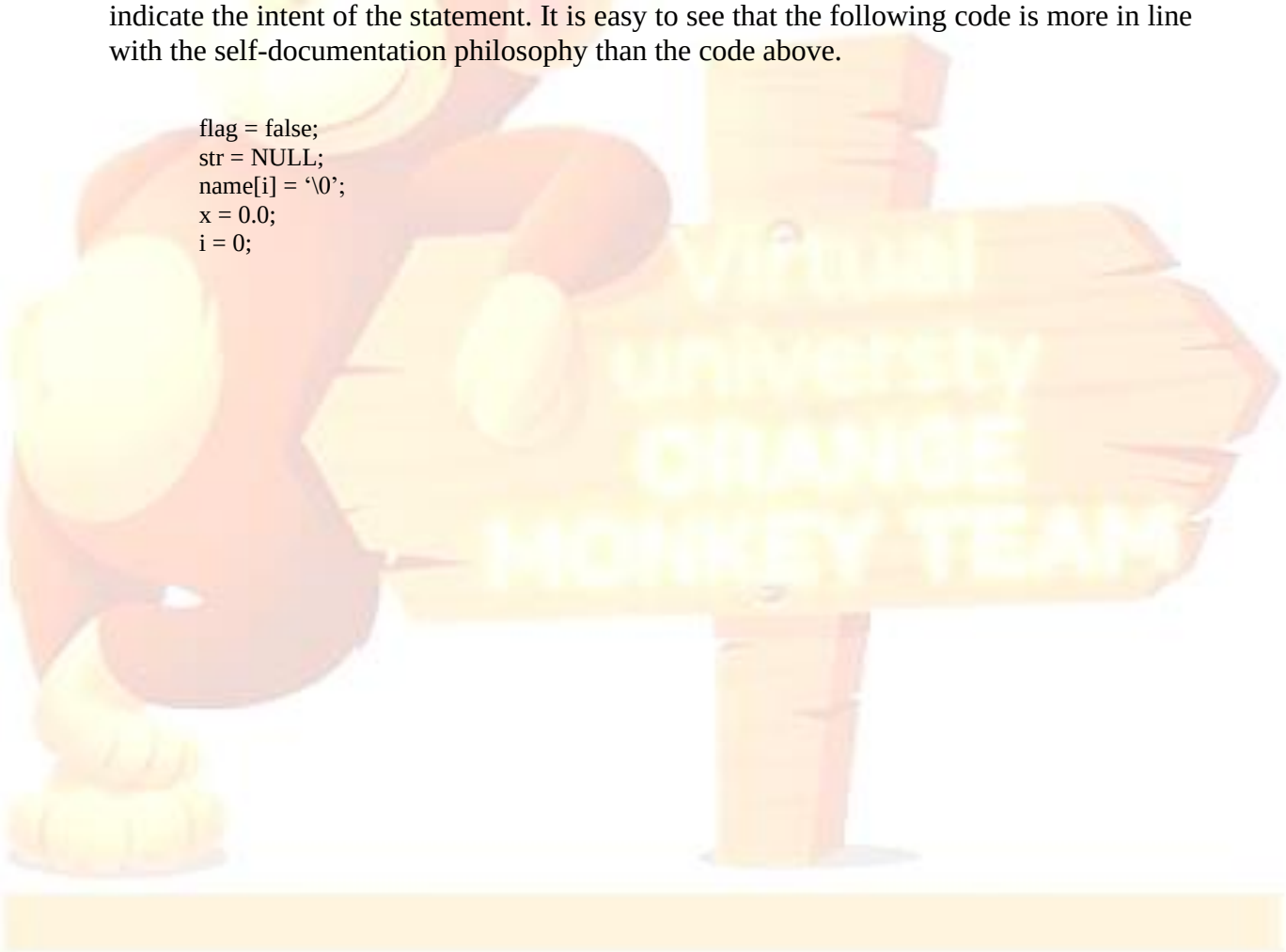
Use (or abuse) of Zero

The number 0 is the most abused symbol in programs written in C or C++. One can easily find code segment that 0 in a fashion similar to the examples below in almost every C/C++ program.

```
flag = 0;          // flag is boolean
str = 0;           // str is string
name[i] = 0;       // name is char array
x = 0;             // x is floating pt
i = 0;             // i is integer
```

This is a legacy of old style C programming. It is much better to use symbols to explicitly indicate the intent of the statement. It is easy to see that the following code is more in line with the self-documentation philosophy than the code above.

```
flag = false;
str = NULL;
name[i] = '\0';
x = 0.0;
i = 0;
```



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Lecture No. 32

Clarity through modularity

As mentioned earlier, abstraction and encapsulation are two important tools that can help in managing and mastering the complexity of a program. We also discussed that the size of individual functions plays a significant role in making the program easy or difficult to understand. In general, as the function becomes longer in size, it becomes more difficult to understand. Modularity is a tool that can help us in reducing the size of individual functions, making them more readable. As an example, consider the following selection sort function.

```
void selectionSort(int a[], int size)
{
        int i, j;
        int temp;
        int min;

        for (i = 0; i < size-1; i++){
                min = i;
                for (j = i+1; j < size; j++){
                        if (a[j] < a[min])
                                min = j;
                }
                temp = a[i];
                a[i] = a[min];
                a[min] = temp;
        }
}
```

Although it is not very long but we can still improve its readability by breaking it into small functions to perform the logical steps. The modified code is written below:

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```
void swap(int &x, int &y)
{
        int  temp;
        temp = x;
        x = y;
        y = temp;
}

int minimum(int a[], int from, int to)
{
        int i;
        int min;
        min = a[from];
        for (i = from; i <= to; i++){
                if (a[i] < a[min])
                        min = i;
        }
        return min;
}

void selectionSort(int a[], int size)
{
        int min;
        int i;
        for (i = 0; i < size; i++){
                min = minimum(a, i, size –1);
                swap(a[i], a[min]);
        }
}
```

It is easy to see that the new selectionSort function is much more readable. The logical steps have been abstracted out into the two functions namely, minimum and swap. This code is not only shorter but also as a by product we now have two functions (minimum and swap) that can be reused.

Reusability is one of the prime reasons to make functions but is not the only reason. Modularity is of equal concern (if not more) and a function should be broken into smaller pieces, even if those pieces are not reused. As an example, let us consider the quickSort algorithm below.

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```
void quickSort(int a[], int left, int right)
{
        int i, j;
        int pivot;
        int temp;
        int mid = (left + right)/2;
        if (left < right){
                i = left - 1;
                j = right + 1;
                pivot = a[mid];
                do {
                        do i++; while (a[i] < pivot);
                        do j--; while (a[i] < pivot);
                        if (i<j){
                                temp = a[i];
                                a[i] = a[j];
                                a[j] = temp;
                        }
                } while (i < j);
                temp = a[left];
                a[left] = a[j];
                a[j] = temp;

                quickSort(a, left, j);
                quickSort(a, j+1, right);
}
```

This is actually a very simple algorithm but students find it very difficult to remember. If is broken in logical steps as shown below, it becomes trivial.

```
void quickSort(int a[], int left, int right)
{
        int p;
        if (left < right){
                p = partition(a, left, right);
                quickSort(a, left, p-1);
                quickSort(a, p+1, right);
        }
}


int partition(int a[], int left, int right)
{
        int i; j;
        int pivot;
        i = left + 1;
        j = right;
        pivot = a[left];
        while(i < right && a[i] < pivot) i++;
        while(j > left && a[j] >= pivot) j++;
        if(i < j)
                swap(a[i], a[j]);
        swap(a[left], a[j]);
        return j;
}
```


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Short circuiting || and &&

The logical and operator, &&, and logical or operators, ||, are special due to the C/C++ short circuiting rule, i.e. a || b and a && b are short circuit evaluated. That is, logical expressions are evaluated left to right and evaluation stops as soon as the final truth value can be determined.

Short-circuiting is a very useful tool. It can be used where one boolean expression can be placed first to "guard" a potentially unsafe operation in a second boolean expression. Also, time is saved in evaluation of complex expressions using operators || and &&. However, a number of issues arise if proper attention is not paid.

Let us look at the following code segment taken from a commercially developed software for a large international bank:

```
struct Node {
        int data;
        Node * next;
};


Node *ptr;
...

while (ptr->data < myData && ptr != NULL){
        // do something here
}
```

What's wrong with this code?

The second part of condition, ptr != NULL, is supposed to be the guard. That is, if the value of the pointer is NULL, then the control should not enter the body of the while loop otherwise, it should check whether ptr->data < myData or not and then proceed accordingly. When the guard is misplaced, if the pointer is NULL then the program will crash because it is illegal to access a component of a non-existent object. This code is rewritten as follows. This time the short-circuiting helps in achieving the desired objective which would have been a little difficult to code without such help.

```
while (ptr != NULL && ptr->data < myData){
        // do something here
}
```


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Operand Evaluation Order and Side Effects

A side effect of a function occurs when the function, besides returning a value, changes either one of its parameters or a variable declared outside the function but is accessible to it. That is, a side effect is caused by an operation that may return an explicit result but it may also modify the values stored in other data objects. Side effects are a major source of programming errors and they make things difficult during maintenance or debugging activities. Many languages do not specify the function evaluation order in a single statement. This combined with side effects causes major problems. As an example, consider the following statement:

```
c = f1(a) + f2(b);
```

The question is, which function (f1 or f2) will be evaluated first as the C/C++ language does not specify the evaluation order and the implementer (compiler writer) is free to choose one order or the other. The question is: does it matter?

To understand this, let's look at the definition of f1 and f2.

```
int f1(int &x)
{
        x = x * 2;
        return x + 1;
}

int f2(int &y)
{
        y = y / 2;
        return y - 1;
}
```

In this case both f1 and f2 have side effects as they both are doing two things - changing the value of the parameter and changing the value at the caller side. Now if we have the following code segment,

```
a = 3;
b = 4;

c = f1(a) + f2(b);
```

then the value of a, b, and c would be as follows:

```
a = 6
b = 2
c = 8
```


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So far there seem to be any problem. But let us now consider the following statement:

```
c = f1(a) + f2(a);
```

What will be the value of a and c after this statement?

If f1 is evaluated before f2 then we have the following values:

```
a = 3
c = 9 // 7 + 2
```

On the other hand, if f2 is evaluated before f1 then, we get totally different results.

```
a = 2
c = 3 // 3 + 0
```


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Lecture No. 33

Common Coding mistakes

Following is short list of common mistakes made due to side-effects.

1.
```
array[i++] = i;
```
If i is initially 3, array[3] might be set to 3 or 4.

2.
```
array[i++] = array[i++] = x;
```
Due to side effects, multiple assignments become very dangerous. In this example, a whole depends upon when i is incremented.

3.
“,” is very dangerous as it causes side effects. Let’s look at the following statement:
```
int i, j = 0;
```
Because of the syntax, many people would assume that i is also being initialized to 0, while it is not. Combination of , and = -- is fatal. Look at the following statement:
```
a = b, c = 0;
```
A majority of the programmers would assume that all a, b, and c are being initialized to 0 while only c is initialized and a and b have garbage values in them. This kind of overlook causes major programming errors which are not caught easily and are caused only because there are side effects.

Guidelines

If the following guidelines are observed, one can avoid hazards caused by side effects.

1. never use “,” except for declaration
2. if you are initializing a variable at the time of declaration, do not declare another variable in the same statement
3. never use multiple assignments in the same statement
4. Be very careful when you use functions with side effects – functions that change the values of the parameters.
5. Try to avoid functions that change the value of some parameters and return some value at the same time.

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Performance

In many cases, performance and maintainability are at odds with one another. When planning for performance, one should always remember the 80/20 rule - you spend 80 percent of your time in 20 percent of the code. That is, we should not try to optimize everything. The proper approach is to profile the program and then identify bottlenecks to be optimized. This is similar to what we do in databases – we usually normalize the database to remove redundancies but then partially de-normalize if there are performance issues.

As an example, consider the following. In this example a function isspam is profiled by calling 10000 times. The results are shown in the following table:

sec	%	Cum%	cycles	instructions	calls	function
			990000	10000	00	strchr
			80000	6000	00	strncmp
			0000	0000	00	strstr
			5559	2213	35	strlen
			000	000		isspam
11 other functions with insignificant performance overhead						

The profiling revealed that most of time was spent in strchr and strncmp and both of these were called from strstr.

When a small set (a couple of functions) of functions which use each other is so overwhelmingly the bottleneck, there are two alternatives:

1. use a better algorithm
2. rewrite the whole set

In this particular case strstr was rewritten and profiled again. It was and found out that although it was much faster but now 99.8% of the time was spent in strstr.

The algorithm was rewritten and restructured again by eliminating strstr, strchr, and strncmp and used memcmp. Now memcmp was much more complex than strstr but it gained efficiency by eliminating a number of loops and the new results are as shown:

sec	%	Cum%	cycles	instructions	calls	function
			0000	90000	000	memcmp
			0000	0000		isspam
			4	3		strlen

strlen now went from over two million calls to 652.

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Many details of the execution can be discovered by examining the numbers. The trick is to concentrate on hot spots by first identifying them and then cooling them. As mentioned earlier, most of the time is spent in loops. Therefore we need to concentrate on loops.

As an example, consider the following:

```
for (j = i; j < MAX_FIELD; j++)
   clear(j);
```

This loop clears field before each new input is read. It was observed that it was taking almost 50% of the total time. On further investigation it was found out that MAX_FIELD was 200 but the actual fields that needed to be cleared were 2 or 3 in most cases. The code was subsequently modified as shown below:

```
for (j = i; j < maxField; j++)
   clear(j);
```

This reduced the overall execution time by half.



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Lecture No. 34

Portability

Many applications need to be ported on to many different platforms. As we have seen, it is pretty hard to write error free, efficient, and maintainable software. So, if a major rework is required to port a program written for one environment to another, it will be probably not come at a low cost. So, we ought to find ways and means by which we can port applications to other platforms with minimum effort. The key to this lies in how we write our program. If we are careful during writing code, we can make it portable. On the other hand if we write code without portability in mind, we may end-up with a code that is extremely hard to port to other environment. Following is brief guideline that can help you in writing portable code.

Stick to the standard

1. Use ANSI/ISO standard C++
2. Instead of using vendor specific language extensions, use STL as much as possible

Program in the mainstream

Although C++ standard does not require function prototypes, one should always write them.

```
double sqrt();              // old style acceptable by ANSI C

double sqrt(double);        // ANSI – the right approach
```

Size of data types

Sizes of data types cause major portability issues as they vary from one machine to the other so one should be careful with them.

```
int i, j, k;

…
j = 20000;
k = 30000;

i = j + k;
```


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```
// works if int is 4 bytes
// what will happen if int is 2 bytes?
```


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Order of Evaluation

As mentioned earlier during the discussion of side effects, order of evaluation varies from one implementation to other. This therefore also causes portability issues. We should therefore follow guidelines mentioned in the side effect discussion.

Signedness of char

The language does not specify whether char is signed or unsigned.

```
char c;
        // between 0 and 255 if unsigned
        // -128 to 127 if signed

c = getchar();
if (c == EOF) ??

        // will fail if it is unsigned
```

It should therefore be written as follows:

```
int c;
c = getchar();
if (c == EOF)
```

Arithmetic or Logical Shift

The C/C++ language has not specified whether right shift >> is arithmetic or logical. In the arithmetic shift sign bit is copied while the logical shift fills the vacated bits with 0. This obviously reduces portability.

Interestingly, Java has introduced a new operator to handle this issue. >> is used for for arithmetic shift and >>> for logical shift.

Byte Order and Data Exchange

The order in which bytes of one word are stored is hardware dependent. For example in Intel architecture the lowest byte is the most significant byte while in Motorola architecture the highest byte of a word is the most significant one. This causes problem when dealing with binary data and we need to be careful while exchanging data between to heterogeneous machines. One should therefore only use text for data exchange. One should also be aware of the internationalization issues and hence should not assume ASCII as well as English.

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Alignment

The C/C++ language does not define the alignment of items within structures, classes, an unions. data may be aligned on word or byte boundaries. For example:

```
struct X {
        char c;
        int i;
};
```

address of i could be 2, 4, or 8 from the beginning of the structure. Therefore, using pointers and then typecasting them to access individual components will cause all sorts of problems.

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Bit Fields

Bit fields allow the packing of data in a structure. This is especially useful when memory or data storage is at a premium. Typical examples:

- Packing several objects into a machine word. *e.g.* 1 bit flags can be compacted -- Symbol tables in compilers.
- Reading external file formats -- non-standard file formats could be read in. *E.g.* 9 bit integers.

C lets us do this in a structure definition by putting :*bit length* after the variable. *i.e.*

```
struct packed_struct {
        unsigned int f1:1;
        unsigned int f2:1;
        unsigned int f3:1;
        unsigned int f4:1;
        unsigned int type:4;
        unsigned int funny_int:9;
} pack;
```

Here the `packed_struct contains 6 members: Four 1 bit` *flags* `f1..f3, a 4 bit type and a 9 bit funny_int.`

C automatically packs the above bit fields as compactly as possible, provided that the maximum length of the field is less than or equal to the integer word length of the computer. If this is not the case then some compilers may allow memory overlap for the fields whilst other would store the next field in the next word.

Bit fields are a convenient way to express many difficult operations. However, bit fields do suffer from a lack of portability between platforms:

- integers may be signed or unsigned
- Many compilers limit the maximum number of bits in the bit field to the size of an `integer` which may be either 16-bit or 32-bit varieties.
- Some bit field members are stored left to right others are stored right to left in memory.
- If bit fields too large, next bit field may be stored consecutively in memory (overlapping the boundary between memory locations) or in the next word of memory.

Bit fields therefore should not be used.

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Lecture No. 35

Exception handling

Exception handling is a powerful technique that separates error-handling code from normal code. It also provides a consistent error handling mechanism. The greatest advantage of exception handling is its ability to handle asynchronous errors.

The idea is to raise some error flag every time something goes wrong. There is a system that is always on the lookout for this error flag. Third, the previous system calls the error handling code if the error flag has been spotted. The raising of the imaginary error flag is simply called *raising* or *throwing* an error. When an error is thrown the overall system responds by *catching* the error. Surrounding a block of error-sensitive code with exception handling is called *trying* to execute a block. The following code segment illustrates the general exception handling mechanism.

```
try {
___...
___...
___throw Exception()
___...
___...
} catch( Exception e )
{
___...
___...
}
```

One of the most powerful features of exception handling is that an error can be thrown over function boundaries. This allows programmers to put the error handling code in one place, such as the *main*-function of your program.

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Exceptions and code complexity

A number of invisible execution paths can exist in simple code in a language that allows exceptions. The complexity of a program may increase significantly if there are exceptional paths in it. Consider the following code:

```
String EvaluateSalaryAnadReturnName( Employee e)
{
  if (e.Title() == "CEO" || e.Salary() > 10000)
  {
        cout << e.First() << " " << e.Last() << " is overpaid" << endl;
  }
  return e.First() + " " + e.Last();
}
```

Before moving any further, let's take the following assumptions:

1. Different order of evaluating function parameters are ignored.
2. Failed destructors are ignored
3. Called functions are considered atomic
    a. for example: e.Title() could throw for several reasons but all that matters for this function is whether e.Title() results in an exception or not.
4. To count as different execution paths, an execution path must be made-up of a unique sequence of function calls performed and exited in the same way.

Question: How many more execution paths are there?

Ans: 23. There are 3 non-exceptional paths and 20 exceptional paths.

The non-exceptional paths:

The three non-exceptional paths are enumerated as below:

1. if (e.Title() == "CEO" || e.Salary() > 10000)
    - if e.Title() == "CEO" is true then the second part is not evaluated and e.Salary() will not be called.
    - cout will be performed

2. if e.Title() != "CEO" and e.Salary() > 10000
    - both parts of the condition will be evaluated
    - cout will be performed.

3. if e.Title() != "CEO" and e.Salary() <= 10000
    - both parts of the condition will be evaluated
    - cout will not be performed.

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Exceptional Code Paths

The 20 exceptional code path are listed below.

1. String EvaluateSalaryAnadReturnName(Employee e)

The argument is passed by value, which invokes the copy constructor. This copy operation might throw an exception.

2. if (e.Title() == “CEO” || e.Salary() > 10000)

e.Title() might itself throw, or it might return an object of class type by value, and that copy operation might throw.

3. if (e.Title() == “CEO” || e.Salary() > 10000)

Same as above.

4. if (e.Title() == “CEO” || e.Salary() > 10000)

To match a valid ==() operator, the string literal may need to be converted to a temporary object of class type and that construction of the temporary might throw.

5. if (e.Title() == “CEO” || e.Salary() > 10000)

Same as above.

6. if (e.Title() == “CEO” || e.Salary() > 10000)

operator ==() might throw.

7. if (e.Title() == “CEO” || e.Salary() > 10000)

Same as above.

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8. if (e.Title() == “CEO” || e.Salary() > 10000)

Same as above.

9-13 cout << e.First() << “ “ << e.Last() << “ is overpaid” << endl;

As per C++ standard, any of the five calls to << operator might throw.

14-15 cout << e.First() << “ “ << e.Last() << “ is overpaid” << endl

similar to 2 and 3.

16-17 return e.First() + “ “ + e.Last();

similar to 14-15.

18-19 return e.First() + “ “ + e.Last();

similar to 6,7, and 8.

20. return e.First() + “ “ + e.Last();

similar to 4.

Summary:

- A number of invisible execution paths can exist in simple code in a language that allows exceptions.
- Always be exception-aware. Know what code might emit exceptions.

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The Challenge:
Can we make this code exception safe and exception neutral? That is, rewrite it (if needed) so that it works properly in the presence of an exception and propagates all exceptions to the caller?

Exception-Safety:
A function is exception safe if it might throw but do not have any side effects if it does throw and any objects being used, including temporaries, are exception safe and clean-up there resources when destroyed.

Exception Neutral:
A function is said to be exception neutral if it propagates all exceptions to the caller.

Levels of Exception Safety

- Basic Guarantee: Ensures that temporaries are destroyed properly and there are no memory leaks.
- Strong Guarantee: Ensures basic guarantee as well as there is full-commit or roll-back.
- No-throw Guarantee: Ensure that a function will not throw.

Exception-safety requires either no-throw guarantee or basic and strong guarantee.

Does the function satisfy basic guarantee?

Yes. Since the function does not create any objects, in the presence of an exception, it does not leak any resources.

Does the function satisfy strong guarantee?

No. The strong guarantee says that if the function fails because of an exception, the program state must not be change.

This function has two distinct side-effects:

- an overpaid message is emitted to cout.
- A name strings is returned.

As far as the second side-effect is concerned, the function meets the strong guarantee because if an exception occurs the value will never be returned.

As far as the first side-effect is concerned, the function is not exception safe for two reasons:

- if exception is thrown after the first part of the message has been emitted to cout but before the message has been completed (for example if the fourth << operator throws), then a partial message was emitted to cout.
- If the message emitted successfully but an exception occurs in later in the function (for example during the assembly of the return value), then a

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message was emitted to cout even though the function failed because of an exception. It should be complete commit or complete roll-back.

Does the function satisfy no-throw guarantee?

No. This is clearly not true as lots of operations in the function might throw.

Strong Guarantee

To meet strong guarantee, either both side-effects are completed or an exception is thrown and neither effect is performed.

```
// First attempt:

String EvaluateSalaryAnadReturnName( Employee e)
{
        String result = e.First() + " " + e.Last();

        if (e.Title() == "CEO" || e.Salary() > 10000)
        {

                String message = result + " is overpaid\n";
                cout << message;

        }
        return result;
}
```

What happens if the function is called as follows:

```
String theName;
theName = evaluateSalarayAndReturnName(someEmplyee);
```

1. string copy constructor is invoked because the result is returned by value.
2. The copy assignment operator is invoked to copy the result into theName
3. If either copy fails then the function has completed its side-effects (since the message was completely emitted and the return value was completely constructed) but the result has been irretrievable lost.

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Can we do better and perhaps avoid the problem by avoiding the copy?

```
// Second attempt:

String EvaluateSalaryAnadReturnName( Employee e, String &r)
{
        String result = e.First() + " " + e.Last();

        if (e.Title() == "CEO" || e.Salary() > 10000)
        {

                String message = result + " is overpaid\n";
                cout << message;

        }
        r = result;
}
```

Looks better but assignment to r might still fail which leaves us with one side-effect completed and other incomplete.

```
// Third attempt:

auto_ptr<String> EvaluateSalaryAnadReturnName( Employee e)
{
        auto_ptr<String> result = new String(e.First() + " " + e.Last());

        if (e.Title() == "CEO" || e.Salary() > 10000)
        {

                String message = (*result) + " is overpaid\n";
                cout << message;

        }
        return result;();       // rely on transfer of ownership
                                // this can't throw
}
```

We have effectively hidden all the work to construct the second side-effect (the return value), while we ensured that it can be safely returned to the caller using only non-throwing operation after the first side-effect has completed the printing of the message. In this case we know that once the function is complete, the return value will make successfully into the hands of the caller and be correctly cleaned-up in all cases. This is because the aut_ptr semantics guarantee that If the caller accepts the returned value, the act of accepting a copy of the auto_ptr causes the caller to take the ownership and if the caller does not accept the returned value, say by ignoring the

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return value, the allocated string will automatically be destroyed with proper clean-up.

Exception Safety and Multiple Side-effects

It is difficult and some-times impossible to provide strong exception safety when there are two or more side-effects in one function and these side-effects are not related with each other. For example we would not have been able to provide exception safety if there are two output messages, one to cout and the other one to cerr. This is because the two cannot be combined.

When such a situation comes with two or more unrelated side-effects which cannot be combined then the best way to handle such a situation is break it into two separate functions. That way, at least, the caller would know that these are two separate atomic steps.

Summary

1. Providing the strong exception-safety guarantee often requires you to trade-off performance.
2. If a function has multiple un-related side-effects, it cannot always be made strongly exception safe. If not, it can be done only by splitting the function into several functions, each of whose side-effects can be performed atomically.
3. Not all functions need to be strongly exception-safe. Both the original code and attempt#1 satisfy the basic guarantee. For many clients, attempt # 1 is sufficient and minimizes the opportunity for side-effects to occur in the exceptional situation, without requiring the performance trade-off of attempt #3.

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Lecture No. 36

Software Verification and Validation

Software Testing

To understand the concept of software testing correctly, we need to understand a few related concepts.

Software verification and validation

Verification and validation are the processes in which we check a product against its specifications and the expectations of the users who will be using it. According to a known software engineering expert Berry Boehm, verification and validation are

Verification

- Does the product meet system specifications?
- Have you built the product right?

Validation

- Does the product meet user expectations?
- Have you built the right product?

It is possible that a software application may fulfill its specifications but it may deviate from users expectations or their desired behavior. That means, software is verified but not validated. How is it possible? It is possible because during the requirements engineering phase, user needs might not have been captured precisely or the analyst might have missed a major stakeholder in the analysis. Therefore, it is important to verify as well as validate the software product.

Defect

The second major and a very important concept is Defect. A defect is a variance from a desired product attribute. These attributes may involve system specifications well as user expectation. Anything that may cause customer dissatisfaction, is a defect. Whether these defects are in system specifications or in the software products, it is essential to point these out and fix.

Therefore software defect is that phenomenon in which software deviates from its expected behavior. This is non-compliance from the expected behavior with respect to written specifications or the stakeholder needs.

Software and Defect

Software and defects go side by side in the software development life cycle. According to a famous saying by Haliburton, Death and taxes are inevitable. According to Kernighan: Death, taxes, and bugs are the only certainties in the life of a programmer. Software and defects cannot be separated, however, it is important to learn how discovering defects at an appropriate stage improves the software quality. Therefore, software application needs to be verified as well as validated for a successful deployment.

Software Testing

With these concepts, we are in a position to define software testing. Software testing is the process of examining the software product against its requirements. Thus it is a process that involves verification of product with respect to its written requirements and conformance of requirements with user needs. From another perspective, software testing

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is the process of executing software product on test data and examining its output vis-à-vis the documented behavior.

Software testing objective

- The correct approach to testing a scientific theory is not to try to verify it, but to seek to refute the theory. That is to prove that it has errors. (Popper 1965)
- The goal of testing is to expose latent defects in a software system before it is put to use.
- A software tester tries to break the system. The objective is to show the presence of a defect not the absence of it.
- Testing cannot show the absence of a defect. It only increases your confidence in the software.
- This is because exhaustive testing of software is not possible – it is simply too expansive and needs virtually infinite resources.

Successful Test

From the following sayings, a successful test can be defined

"If you think your task is to find problems then you will look harder for them than if you think your task is to verify that the program has none" – Myers 1979.

"A test is said to be successful if it discovers an error" – doctor's analogy.

The success of a test depends upon the ability to discover a bug not in the ability to prove that the software does not have one. As, it is impossible to check all the different scenarios of a software application, however, we can apply techniques that can discover potential bugs from the application. Thus a test that helps in discovering a bug is a successful test. In software testing phase, our emphasis is on discovering all the major bugs that can be identified by running certain test scenarios. However it is important to keep in mind that testing activity has certain limitations.

Limitations of testing

With the help of the following example, we shall see how difficult it may become to discover a defect from a software application.

Example (adapted from Backhouse)

This is a function that compares two strings of characters stored in an array for equality. The tester who has to test this function has devised the following test cases in which different combinations of inputs are tried and their expected behavior is documented. From the look of this table, it seems that almost all major combinations of inputs have been documented.

Inputs and Expected Outputs

A	b	Expected result
"cat"	"dog"	False
""	""	True
"hen"	"hen"	True
"hen"	"heN"	False
" "	""	False

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“”	“ball”	False
“cat”	“”	False
“HEN”	“hen”	False
“rat”	“door”	False
“ ”	“ ”	True

Results of testing

The tester runs all the above-mentioned test cases and function returns the same results as expected. But it is still not correct. What can be a problem To analyze the problem, lets look at the code of this string equal routine

Code of the Function

```
bool isStringsEqual(char a[], char b[])
{
   bool result;
   if (strlen(a) != strlen(b))
      result = false;
   else {
      for (int i =0; i < strlen(a); i++)
         if (a[i] == b[i])
            result = true;
         else result = false;
   }
   return result;
}
```

Analysis of the code

It passes all the designated tests but fails for two different strings of same length ending with the same character. For example, “cut” and “rat” would results in true which is not correct.

The above-mentioned defect signifies a clear limitation of the testing process in discovering a defect which is not very frequent. However, it should be noted from this example that a tester cannot generate all possible combinations of test cases to test an application as the number of scenarios may become exhaustive.

Testing limitations

- In order to prove that a formula or hypothesis is incorrect all you have to do to show only one example in which you prove that the formula or theorem is not working.
- On the other hand, million of examples can be developed to support the hypothesis but this will not prove that it is correct.
- These examples only help you in coming up with a hypothesis but they are not proves by themselves and they only enhance your comfort level in that particular hypothesis or in this particular case, in your piece of software.
- You cannot test a program completely because:
  - the domain of the possible inputs is too large to test.
  - there are too many possible paths through the program to test.
- According to Discrete Mathematics

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- To prove that a formula or hypothesis is incorrect you have to show only one example.
- To prove that it is correct any numbers of examples are insufficient. You have to give a formal proof of its correctness.

Test Cases and Test Data

In order to test a software application, it is necessary to generate test cases and test data which is used in the application. Test cases correspond to application functionality such that the tester writes down steps which should be followed to achieve certain functionality. Thus a test case involves

- Input and output specification plus a statement of the function under test.
- Steps to perform the function
- Expected results that the software application produces

However, test data includes inputs that have been devised to test the system.

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CS504-Software Engineering-I

FINAL TERM MCQS

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ALL answers are verified if found any mistake then Correct ACCORDINGLY

1. Thin Client Model is one of the configurations of_____Model.
   a. Data-centered
   b. Client Server page130
   c. Layered
   d. Reference
2. STL Stands for_____
   a. Standard Template Library Page 138
   b. Standard Type Link
   c. Standard Tempo Line
   d. Standard Type Link
3. The three basic principles that guide maintainability are:
   (a) Clarity
   (b) Enhancement
   (c) Flexibility
   (d) Simplicity
   a. (a),(b) and (c)
   b. (a),(c) and (d) page 149
   c. (b),(c) and (d)
   d. All of above
4. Goto statements violate the idea of
   a. object oriented code
   b. structured code page 164
   c. control structure
   d. repetition structure
5. A function should not be larger in any case and should not exceed_____in length
   a. Half page
   b. One page page 150
   c. Two pages
   d. Three Pages

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6. Data Centered Architectural Style is also called_____
   a. **Repository model page 129**
   b. Client Server model
   c. Sub System model
   d. Reference model
7. Global variables in C++ should always be referred to by using the
   a. **::operator page 153**
   b. :operator
   c. Without an operator
   d. None of the given
8. There are____layers in OSI reference model.
   a. 5
   b. 6
   c. **7 page 134**
   d. 8
9. Dynamic process model shows the process _____of theSystem.
   a. Components
   b. Objects
   c. **Structure page 129**
   d. Linkage
10. Which of the following shows a commented statement in C++
   a. # Ans = first + second
   b. **// Ans = first + second page 162**
   c. \\ Ans = first + second
   d. /# Ans = first + second
11. _____Architectural model is widely used in Mainframe application.
   a. Client-Server Model
   b. **Repository Model page 130**
   c. Filter Model
   d. Layered Model

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12. Variables should be initialized where they are_____and they should be declared in the_____scope possible.
    a. defined and smallest
    b. declared and medium
    c. defined and medium
    d. declared and smallest page 161

13. Identifier names also play a significant role in enhancing the_____ of a program,
    a. Writ ability
    b. Readability page 151
    c. Reliability
    d. All of the given choices

15. MVC stands for_____
    a. Model View Controller page 140
    b. Modern View Center
    c. Model View Center
    d. Modern View Controller

16. Which of the following is not a possible server in client server environment?
    a. Database Server
    b. Transaction Server
    c. File Server
    d. Time Server page 132

17. In case of using unrelated operators in a single expression,____would be the best choice to prevent the logical errors.
    a. Comments
    b. Indents
    c. Parenthesis page 166
    d. Short Cuts

18. In case of header files, construction is to avoid___ errors, the construction should appear in the top of the file (before the file header).
    a. Compilation page 160
    b. run time
    c. logical
    d. All of the given

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19. Which one of these represents the Krutchen's 4+1 architectural view model
    a. **Logical view, Process view. Physical view. Development view. Use case view**
    b. Logical view. Dynamic view. Physical view. Development view. Use case view
    c. Logical view. Process view, Physical view. Development view. Sequence view
    d. Dynamic view, Process view. Physical view, Development view. Use case view
20. Client server is a_____system model.
    a. Integrated
    b. Distributed page 130
    c. Heterogeneous
    d. Homogeneous
21. Vertical partitioning is also known as_____
    a. Balancing
    b. Mutating
    c. Parallelizing
    d. Factoring page 135
22. "is" prefix should be used for_____variables and methods.
    a. General
    b. Boolean page 152
    c. Constant
    d. None of the given
23. Thin client model places a heavy processing load on____
    a. Only server
    b. Only network
    c. Both server and network page 133
    d. Neither server not network

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24. C++ header files should have the extension____.Source files can have the extension_____
    a. .c and .h
    b. .cc and .h
    c. .h and.cpp page 158
    d. All of the given
25. Which of following is among ten things, which the basic template of GOF design pattern includes.
    a. IQ
    b. Context page 142
    c. Performance
    d. Requirements
26. Vertical partitioning divides the architecture application from a____making perspective.
    a. Decision page 135
    b. Design
    c. Conclusion
    d. Move
27. Issues like system performance, availability, scalability, and security are analyzed in_____
    a. Logical View
    b. Physical View page 127
    c. Code View
    d. Concurrency View
28. Class variables should never be declared public. Public variables violate whichof the following:
    a. Information hiding
    b. Encapsulation
    c. Information hiding and Encapsulation page 161
    d. None of the give
29. Charles Simonyi first discussed Hungarian Notation, He was of_____
    a. Microsoft page152
    b. IBM
    c. Dell
    d. Cisco

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30. Patterns are devices that allow programs to shareknowledge about their_____
    a. Code
    b. Design page 137
    c. Analysis
    d. Implementation
31. Process view in Krutchen's 4+1 architectural view model captures_____
    a. Object model of the design
    b. Concurrency and synchronization aspects of the design page 122
    c. Mapping(s) of the software onto the hardware and reflects its distributed aspect
    d. Static organization of the software in its development environment
32. In the N-Tire architecture , the idea is to enhance scalability and_____by disturbing both the date and the application using multiple server machines
    a. Usability
    b. Performance page 134
    c. Interoperability
    d. Integrity
34. Names representing methods and functions should be_____and written in mixed case starting with____case
    a. Noun__lower
    b. Verb___lower page 153
    c. Noun___upper
    d. Verb___upper
35. Floating point constant should always be written with decimal point and at least
    a. One decimal page 164
    b. Two decimal
    c. Three decimal
    d. none of the given
36. Static structural model shows the major system _____
    a. Leaks
    b. Components page 129
    c. Activates

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37. Objectives Distributing the responsibilities to different subsystems so that we get a software system which is easy to maintain, is called_____ the architecture
    a. Subtracting
    b. Partitioning page 138
    c. Cloning
    d. Balancing
38. It ensures that a class only has one instance and provides a global point of access to it
    a. Singleton pattern page 145
    b. Observer pattern
    c. Real pattern
    d. None of the given
39. N-tier architecture stems from the struggle to find a___between the fat-client architecture and the thin client architecture
    a. Concurrency
    b. Distribution point
    c. Middle ground page 134
    d. Similarity
40. In client server model, the application is modeled as a set of that areprovided by servers.
    a. Request
    b. Protocols
    c. Service page 130
    d. Requirements
41. Maintainability and portability issues are discussed in _____
    a. Functional view
    b. Physical view
    c. Code view Page 126
    d. Logical view

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42. In case of a file server, client request selected records from a___and the servertransmits recorded to client over the network
    a. Local memory
    b. Network
    c. Database
    d. File Page132
43. Logical view in Krutchen's 4+1 architectural view model captures_____.
    a. Concurrency and synchronization aspects of the design
    b. Mapping(s)of the software onto the hardware and reflect its distributedaspect
    c. Object model of the design
    d. Static organization of the software in its development environment
44. When programmers use shortcuts and cryptic codes in their program_____ becomes a major problem.
    a. Maintenance
    b. Reusability
    c. Availability
    d. Readability
45. Which one is correct?
    a. double total= 0.5;
    b. double total = .5;
    c. double total = .50;
    d. all of the given
46. "Description of communicating objects and classes that are customized to solvea general design problem in a particular context" is called _.
    a. Design Pattern
    b. System Pattern
    c. System Design
    d. Design Method
47. Development view in Krutchen's 4+1 architectural view model captures
    a. Object model of the design
    b. Concurrency and synchronization aspects of the design

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c. Mapping(s) of the software onto the hardware and reflects its distributedaspect

d. Static organization of the software in its development environment

48. Client server model tries to_____data and processing.

a. Distribute

b. Merge

c. Clone

d. Proceed

49. Which indent size eliminates the chance of code lines splitting?

a. 2

b. 4

c. 3

d. 6

50. Which one of these is a correct short cut used by the programmer?

a. x*=a

b. x=*a

c. x*a=

d. =a*x

51. _____was the first pure Object Oriented language in which observer patternwas used in implementing its Model View Controller pattern

a. Smalltalk

b. PASCAL

c. JAVA

d. C++

52. Identify the TRUE statement

a. Portability effects Security

b. Size of data types vary from one machine to other

c. Size of control structures vary from one machine to other

d. None of the given options

53. 1) x = (a +2 > 3)? a : a-1;
2) if((a + 2)> 3)
x =
a;else
x = a -1;

a. Statement (1) is more complex than (2)

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b. Statement (2) is more complex than (1)
c. Both statements are very complex
d. None of the given options

54. which of the following statements are same in output1) a = a >> 2
2) a = a/4
3) a = a * 2
a. (1) and (3) only
b. (2) and (3) only
c. (1) and (2) only page 168
c. All produce the same result

55. Code should not be:
a. Commented page 130
b. Indented
c. aligned

56. If (!(block < active Block))is equivalent to:
a. if (block < activeBlock))
b. if ((block == activeBlock))
c. if ((block >= activeBlock)) page 163
d. None of the given

57. The form for (;;) should be used for
a. Nested loop
b. **Empty loop page 159**
c. More than 1000 iteration
d. Less than 1000 iterations

58. using proper parenthesis normally makes the code
a. easy to read
b. easy to understand page 167
c. less ambigous
d. All of the given option

59. The clients and servers, in a client server environment, are connected through _____.

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a. Framework
b. Interface
c. Middleware page 129
d. Groupware

60. _____provides a unified interface to a set of intelligence in a sub-system
a. Observer pattern
b. Singleton pattern
c. Façade pattern page 143
c. All of the above

61. Complex expressions:
a. Make the code easy to modify
b. Make the code difficult to modify page 164
c. Make the code easy to understand
d. Does not affect understandability

62. Bring-in system's a view and define, from the system's perspective, the Software functionally the developers must build.
a. Functional Requirements
b. Non-Functional Requirements
c. User Requirements
d. Business Requirements

63. Which of the following is used for multi-level commenting?
a. // Comment
b. /*Comment*/
c. (Comment)
d. */Comment/*

64. Which of the following is NOT an objective for building an analysis model?
a. Define set of software requirements
b. Describe customer requirements
c. Develop an abbreviated solution for the problem
d. Establish basis for software design

65. Consider the following scenario:
"Student logins the system and checks his/her lecture schedule"
Keeping in mind the use case diagram, and above scenario the one of the post conditions might be:
a. Students should have a login

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b. System should be available
c. Students should log-off after checking schedule
d. Lecture schedule should be displayed date wise

66. OOA is intended to define------, their relationships, and their behavior.
a. Variables
b. Classes
c. Objects
d. Subjects

67. One of the most powerful features of exception handling is that an error can be------over function boundaries.
a. Thrown
b. Called back
c. Caught
d. Sent

68. Inspections can check conformance with a specification but not conformance with the------requirements.
a. Customer's real
b. Developer
c. Tester
d. Manger

69. For inspections, ------are prepared that contain information regarding defects.
a. Tables
b. Checklists
c. Lists
d. Farms

70. The first "bug" was actually a moth, which flew through an open window and into one of the Mark -------'s relays.
a. II
b. I
c. III
d. IV

71. Holistic medicine, concerns itself with the state of the body as a whole, not the-----that is currently attacking it.
a. Target
b. Reason
c. Disease
d. Source

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72. What factor has no precipitation in more sophisticated and complex computer-based systems?
    a. Vast use of personal computers
    b. Vast increases in computer memory and storage capacity.
    c. Greater variety of exotic input/output options.
    d. Profound changes in computer architectures.
73. What types of errors are missed by black-box testing and can be uncovered by white-box testing?
    a. Runtime errors
    b. Logic errors
    c. Performance errors
    d. Input errors
74. When large amount of data is to be shared, repository model is used. This model has been extensively used in the--------based application.
    a. Mainframe
    b. Super Computers
    c. Personal Computers
    d. Real time
75. A (n) ----------is a variance from a desired product attribute.
    a. Error
    b. Exception
    c. Defect
    d. Mistake
76. Performance &usability are example of------requirements.
    a. Business
    b. Functional
    c. Non-Functional
    d. User
77. Camel Case is now the official convention for file names and identifiers in the-------programming language.
    a. C#
    b. C++
    c. Java
    d. Visual Basic. Net
78. Comments should be indented relative to their position in the-------.
    a. Code
    b. Design

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c. Analysis
d. Requirements

79. One of the guidelines to avoid common mistakes is to never use-------except foe declaration.
    a. ,
    b. :
    c. =
    d. II
80. When planning for performance, one should always remember the-------rule.
    a. 80/20
    b. 70/20
    c. 60/20
    d. 100/20
81. Behavioral class pattern uses-------relationship to distribute behavior between classes.
    a. Composition
    b. Aggregation
    c. Association
    d. Inheritance
82. Software crisis appeared in-------.
    a. Early 50s
    b. Early 60s
    c. Early 70s
    d. Early 80s
83. In C++ or java, named constants are written in all uppercase letters with-----to separate words.
    a. Semicolon;
    b. Underscore_
    c. Colon:
    d. Dot.
84. Indentation larger than------makes deeply nested code difficult to read and increases the chance that the lines must be split.
    a. 2
    b. 4
    c. 3
    d. 1
85. Which phase of software development lifecycle is considered most expensive?

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a. Requirement Gathering
b. System Design
c. Maintenance
d. Development

86. A software requirement document describes all the-----provided by the system along with the constraints under which it must operate.
a. Conditions
b. Services
c. Tasks
d. Actions

87. ------and functional specification documents are produced in requirement analysis phase.
a. Feasibility report
b. Requirement definition
c. Requirement specification
d. Design definition

88. During the program execution most of the time is spent on-------.
a. Conditional Structure
b. Loops
c. Arrays
d. Functions

89. Many applications need to be ported on to many different-------.
a. Platforms
b. Language
c. Compilers
d. Software

90. There are-------steps involved in identification of structure while deriving object model.
a. Two
b. Three
c. Four
d. Five

91. The technique which is used to separate error-handling code from normal code is called--------.
a. Exceptional handling
b. Function handling
c. Code handling

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d. Variable handling

92. In order to show the presence of a defect, a-----breaks the system.
    a. Tester
    b. Developer
    c. Requirement engineer
    d. Designer
93. Architectural model proposed by Clements et. al. contains------views.
    a. 3
    b. 4
    c. 5
    d. 6
94. -------server provides set of applications that enable communication among clients using text, images, bulletin boards, video, etc.
    a. File
    b. Database
    c. Transaction
    d. Groupware
95. -----is result of efforts to find a middle ground between the fat client architecture and the thin-client architecture.
    a. Pipe and filter architecture
    b. N-tier architecture
    c. Three tier architecture
    d. Zero install
96. The process view captures the concurrency and-------aspects of the design.
    a. Synchronization
    b. ASynchronization
    c. Mapping
    d. Data
97. -------involve purely black box testing.
    a. Unit testing, Beta testing
    b. Acceptance testing, Interfacing
    c. Beta testing, Acceptance testing
    d. Integration testing, Interface testing
98. ------is the correct formula for calculating Cyclomatic complexity of a program.
    a. V – N +2
    b. E – V + 2

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c. E – N + 2
d. E + N – 2

99. ---------is not included in test criteria applied in a phase of testing.
a. Functional validity
b. Interface integrity
c. Correctness
d. Programming Logic

100. The nature of software application can be characterized by their information-------.
a. Complexity
b. Content
c. Determinacy
d. Content and determinacy

101. The best way to conduct a requirements validation review is to-------.
a. Examine the system model for errors
b. Have the customer look over the requirements
c. Send them to the design team and see if they have any concerns
d. Use a checklist of questions to examine each requirements

102. Flow charts represents--------.
a. Sequence Activity
b. Random Activity
c. Parallel Activity
d. Shuffle Activity

103. A process which does not take input is called-------.
a. Miracle process
b. Core process
c. Secondary process
d. Zombie process

104. State Transition Diagram is helpful in determining--------.
a. Business understanding
b. Process flow
c. Data store
d. Control flow

105. In a----------, each program module is represented by a rectangular box.
a. Use case diagram
b. Document flow diagram
c. Class diagram

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d. Data flow diagram

106. Which of the following is not a fundamental structured programming construct?
   a. Recursion
   b. Condition
   c. Repetition
   d. Sequence

107. Which writing style is best regarding identifier role in enhancing the readability of a program?
   a. If (Flag==0)
   b. If (Flag==START_NUMBER)
   c. If (Z==START_NUMBER)
   d. If (Z==0)

108. Exception handling is a powerful technique that separates error-handling code from--------code.
   a. Normal
   b. Abnormal
   c. Single
   d. Complex

109. Static analyzers are software tools for--------processing.
   a. Analysis text
   b. Source text
   c. Design text
   d. Maintenance text

110. Testing is an intellectually demanding activity and has a life cycle------to software development.
   a. Equal
   b. Parallel
   c. Sequential
   d. In contrast

111. --------are used to describe flow of data or control in an application.
   a. Code structures
   b. String matchings
   c. Paths
   d. Flow graphs

112. Writing test cases and generating test data are processes that demand------ building capabilities.

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a. Code
b. Domain
c. System
d. Scenario

113. Idea of zero install architecture is to develop a system where no installation on the-----is needed.
a. Client side
b. Server side
c. Client & server side
d. Network

114. There are four basic coding structures: sequence, if statement, case statement, and---------.
a. For loop
b. While loop
c. Switch statement
d. Logical operations

115. The pattern movement became very quiet until------when patterns appeared again at OOPSLA conference.
a. 1987
b. 1988
c. 1962
d. 1995

116. In-------coverage scheme, all possible paths of a program from input instruction to the output instruction are tested.
a. Branch coverage
b. Statement coverage
c. Path coverage
d. System coverage

117. Inspections cannot check-------characteristics.
a. Non-Functional
b. Business
c. User
d. Functional

118. Split lines occur when a statement exceed the-----column limit.
a. 80
b. 90
c. 95

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d. 85

119. STL is a library of reusable components provided by------compilers.
   a. Java
   b. Cobol
   c. Small talk
   d. C++

120. In-------, the responsibility of instantiation of an object is given to some other object.
   a. Behavioral
   b. Structural
   c. Creational
   d. Observer

121. ----------pattern is an example of creational pattern.
   a. Façade
   b. Singleton
   c. Observer
   d. Prototyping

122. ------pattern hides the implementation of the subsystem from clients, making the subsystem easier to use.
   a. Observer
   b. Fecade
   c. Creational
   d. Singleton

123. --------language allows short circuiting.
   a. Pascal
   b. C
   c. C++
   d. ADA

124. Dynamic memory allocation is done from internal memory storage called---.
   a. Cache
   b. Pool
   c. VRam
   d. Boot Memory

125. Context level diagram is also known as-------.
   a. 0-level DFD
   b. 1-level DFD
   c. 2-level DFD

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d. 3-level DFD

126. Unit testing process is done by--------stakeholder of the software.
   a. Project manger
   b. Customer
   c. Software
   d. Developer

127. The number of paths in a program that contains loops tends to---------.
   a. Finite
   b. Infinity
   c. Limited
   d. Countable

128. UML is among different------which are used for documenting the object oriented design.
   a. Reports
   b. Documents
   c. Notations
   d. SRS

129. White box testing is performed to test the _____ of the program
   a. Logic
   b. Structure page 198
   c. Flow
   d. Syntax

130. Some bit field members are stored:
   - left to right
   - right to left
   - in circular array

   a. only (1) is true
   b. Only (ll) is true
   c. Only (lll) is true
   d. e. Both (l) and (II) are true page 183

131. In order to write a portable code which of the following guideline will not be helpful:
   a. Stick to the standards
   b. Program in the mainstream
   c. Size of data types
   d. **Using vendor specific language extensions**

132. _____is qualitative benefit of Unit Testing

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a. Repeatability
b. **Confidence building page 207**
c. Bounded code
d. Cost effectiveness

133. In order to development life cycle. Instead of using vendor specific language extensions, use _____as much as possible
a. STL page 179
b. ANSI
c. ISO
d. CMM

134. In system development life cycle, defects cannot enter into the program during _____ phase.
a. Requirement
b. Design
c. Coding
d. **Implementation page 209**

135. _____and _____ are important short circuiting logical operators.
a. AND b. OR c. NOTd. NOR
a. **a & b pae 173**
b. b&c
c. c&d
d. a & c.

136. _____ testing isolates every part of the program and shows that the individual parts are correct.
a. White Box
b. Black Box
c. **Unit Google**
d. System

137. _____is not a white box testing technique.
a. Statement Coverage
b. Branch Coverage
c. Path Coverage
d. State Transition Coverage page 202

138. Chip level testing for hardware is equivalent to _____
a. White Box
b. Black Box
c. **Unit testing page 207**

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d. System testing

139. For equivalence partitions, we divide the problem in _____ obvious categories.
   a. **Two**
   b. Four
   c. Five
   d. Six

140. Which rewritten form of above line of code is more in line with the self-documentation philosophy than the code above?
   a. x = false
   b. x = NULL
   c. **X = 0.0; page 169**
   d. x = '\0`

141. One of the main reasons to make functions is _____
   a. Reliability
   b. **Reusability page171**
   c. Maintainability
   d. Efficiency

142. 80/20 rule states that:
   a. **you spend 80 percent of your time in 20 percent of the code page 177**
   b. you spend 20 percent of your time in 80 percent of the code
   c. We should try to optimized 80 percent or at least 20 percent of the code
   d. We should try to optimized 20 percent or at least 80 percent of the code

143. _____ identifies which statements in a method or class have been executed.
   a. Method Coverage
   b. **Statement Coverage**
   c. Class Coverage
   d. Branch Coverage

144. In _____ system's internal implementation details are not visible to the tester.
   a. White Box Testing
   b. Gray Box Testing
   c. **Black Box Testing page 198**
   d. Both White Box Testing & Gray Box Testing

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145. Verification and validation are the processes in which we check a product against its _____ and the _____ of the users who will be using it

a. specifications arguments
b. statements, expectations
c. **specifications, expectations** **page 192**
d. statements, arguments

146. Objective of _____ is to show that the program does not work

a. Requirements
b. Design
c. Coding
d. **testing** **page 196**

147. The raising of the imaginary error flag is simply called raising or ___ an error

a. Catching
b. Casting
c. **Throwing** **page 184**
d. Repress

148. Be very careful when you use functions with side effects - functions that change the values of the _____

a. Objects
b. Classes
c. Structures
d. **Variables**

Be very careful when you use functions with side effects – functions that change the values of the parameters. Page 176

149. The number _____ is the most abused symbol in programs written in C or C++.

a. 1
b. 0 page169
c. 2
d. 3

150. _____ is a creative activity while_____ is a destructive activity.

a. **Development, Testing**
b. Testing, Development
c. Validation, Verification
d. Verification, Validation

151. Testing individual components independent of other components is called:

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a. **Unit testing page 189**
b. Module testing
c. Subsystem testing
d. System testing

152. The expressions with logical operators can be evaluated only from_____
a. Right to left
b. **Left to right page173**
c. Top to bottom
d. Bottom to top

153. The complexity of a program may _____if there are exceptional paths in it
a. Decrease
b. **Increase page 185**
c. Remain same
d. Cutback

154. In the switch statement, cases should always end with a _____ statement.
a. Switch
b. Go
c. **Break page 167**
d. Stop

155. Cyclomatic complexity measures the logical complexity____
a. Quantitatively page 205
b. Qualitatively
c. Creditably
d. Worthily

156. The size of _____ plays a significant role in making the program easy or difficult to understand
a. **Function page 170**
b. Object
c. data type
d. none of the above

157. _____depicts programming instructions that do not have branching or any control information
a. **Sequence page 200**
b. IF
c. While
d. Case

158. for (i =0, col = 0; i < 27; i++.j++)

In the above line of the code 27 is representing _____.

a. Real Numbers
b. **Magic Numbers page 167**
c. Constant Numbers
d. Positive Numbers

159. The process that involves verification of product with respect to its written requirements is called _____
a. Maintenance
b. Debugging
c. Development
d. **Testing page 192**

160. If an application fulfills its specifications but deviates from users expectations or their desired behavior. This means, software is verified but not _____
a. **Validated page 192**
b. Corrected
c. Checked
d. Traced

161. If a function changes the value of some other accessible data object along with returning its value after execution is called _____

a. Short Circuiting
b. Modularity
c. **Side Effects page 174**
d. Abstraction

162. if (e.Title() == "CEO" || e. Salary() > 10000)
in the above statement if (e.Title()=="CEO") is TRUE then
a. **Second part is not evaluated page 185**
b. Second part is always evaluated
c. Second part must also be TRUE
d. Second part is not evaluated but e.Salary ()is called

163. When an error is thrown the overall system (on the lookout for this error flag) responds by ____the error

a. Ignoring
b. Casting
c. **Catching page 184**
d. Pass over

164. Testing of collection of modules to discover interfacing problems among interacting modules is called

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a. Unit testing
b. Module testing
c. **Subsystem testing page 198**
d. System testing

165. Constant values used in the program are called
a. Real Numbers
b. **Magic Numbers page 168**
c. Constant Numbers
d. Positive Numbers

166. Bit fields are a convenient way to express many difficult operations. However, bit fields suffer from one problem
a. Lack of usability
b. Lack of security
c. Lack of performance
d. **Lack of portability page 183**

167. Comma (,) is very dangerous because _____
a. Compiler does not recognize this symbol
b. It creates linkage problem
c. **It causes side effects page 176**
d. It does not causes side effects

168. Function testing falls under _____testing technique
a. White Box
b. Black Box
c. **Unit page 207**
d. System

169. The greatest advantage of exception handling is its ability to handle:
a. **Asynchronous errors page 184**
b. Syntax errors
c. Memory errors
d. Control Structure errors

170. After integrating subsystems into a system, and then testing this system as a whole is called
a. Unit testing
b. Component testing
c. Subsystem testing
d. **System testing**

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171. struct packed_struct{

unsigned int
f1:1; } pack;

Here in "packed_struct":

a. value of f1 = 1
b. size of f1 = 1 bit page 183
c. value of f1 should not exceede 1
d. None of given options

172. The C/C++ language has not specified whether _____ is arithmetic or logical.
a. Right shift >> page 181
b. Right shift <<
c. &&
d. ||

173. There are_____ basic coding structures.

a. One
b. Two
c. Three
d. Four page 200

174. In _____ control can take either of several branches.
a. IF
b. While
c. Case statement page 201
d. Sequence

175. _____ cause major portability issues
a. Loops
b. Bugs in code
c. Sizes of data types page 179
d. Conditional Structures

176. A good program must contain _____ infeasible paths.

a. 0 page 206

b. 1
c. N
d. Infinite

177. We do not use flow graphs to _____.
a. Show flow of data

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b. **Describe decisions**
c. Show Control in an application
d. Describe different coding structures

178. The order in which bytes of one word are stored is _____ dependent
   a. **Hardware page 181**
   b. Software
   c. Language
   d. syntax

179. _____ is the starting point, base document for both testing and the development
   a. Software code
   b. Test case document
   c. Software architecture document
   d. **Functional specification document page196**

180. if a major rework is required to translate a program written for one environment to another, it means code is not or less _____
   a. Secure
   b. Bug Free
   c. platform dependent
   d. **Portable page 179**

181. Which of the following is the correct path for the code:

```
If(a%b==0)      1
C=a+b;          2
Cout<<c;        3
```

   a. 1-2-3-1
   b. 1-3-2
   c. 1-2-3
   d. 2-3-1

182. Switch statement is equal to _____ statement.
   a. if
   b. if-else
   c. nested if
   d. **if-else if page 167**

183. Exception handling provides
   a. Portability mechanism

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b. Code Usability mechanism
c. **Error Handling mechanism page 184**
d. Both Portability & Code Usability mechanism

184. Unit Testing will be done by

a. Testers
b. End Users
c. Customer
d. **Developers page 207**

185. Which type of testing is not concerned with how the inputs are transformed into outputs?

a. White Box
b. Gray Box
c. **Black Box page 198**
d. Both White Box & Gray Box

186. Consider the following statement: int a,b=10;
Which of the following is correct?

a. variable "a" is initialized to 10
b. **Variable "b" is initialized to 10**
c. Both variables "a" and "b" are initialized to 10
d. variables cannot be initialized this way

187. Software _____ is the process of examining the software product against its requirements.

a. Testing page 192

b. Debugging
c. Exception handling
d. Fixing

188. The idea behind exception handling is to raise some error flag every time

a. The code compiles
b. The code links

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c. Memory is allocated
d. **Something goes wrong page 184**

189. Bit fields allow the packing of data in a structure, using Bit fields we can:
a. **Read 9 bit integers page 182**
b. Avoid memory leakages
c. Avoid memory overflow
d. Avoid syntax errors

190. _____ is a tool that can help us in reducing the size of individual functions.
a. Inheritance
b. **Modularity page 170**
c. Association
d. Abstraction

191. _____ and _____ are two important tools that helps in managing the program complexity
a. Composition, Inheritance
b. **Abstraction, Encapsulation page 170**
c. Aggregation, Inheritance
d. Modularity, Composition

192. A path through a program which is never traversed for any input data is called
a. Dependent path
b. Independent path
c. **Infeasible path page 206**
d. Feasible path

193. When a small set of functions (which use each other) is so overwhelmingly the bottleneck, there are two alternatives:
a. **use a better algorithm OR re-write the code page 177**
b. debug the code OR place assertions in code
c. remove the functions OR add more functions
d. changed programming language OR compiler at least

194. **The first step in any OOA process model is to_____**
a. build an object-relationship model
b. define collaborations between objects
c. **elicit customer requirements Page603**
d. select a representation language

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195. **A necessary supplement to transform or transaction mapping needed to create a complete architectural design is _____.**
   a. entity relationship diagrams
   b. the data dictionary
   c. **processing narratives for each module Page422**
   d. test cases for each module

196. **All comments should be written in---------**
   a. **English (Page 162)**
   b. French
   c. C++
   d. JAVA

197. **Which test criteria should be applied in a phase of testing?**
   a. functional validity
   b. interface integrity
   c. correctness
   d. **all of the given Page522**

198. **The hardest single part of building a software system is deciding precisely ---------- to build.**
   a. **what (Page 17)**
   b. How
   c. When
   d. Why

199. **In sequence Diagrams objects are organized in a ----------line**
   a. **horizontal (Page 106)**
   b. vertical
   c. horizontal and vertical
   d. none of the above

200. **Every view is potentially Hierarchical in this regard which of the view are Hierarchical?**
   a. Functional View
   b. Development View
   c. C*oncurrency view*
   d. **All of given (Page 125)**

201. **Like analysis models, many different kinds of ----------models are developed**
   a. **Architectural (Page 126)**
   b. System
   c. Design
   d. Logical

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202. **Idea behind zero install architecture is to develop a system where no installation on the ---------- is needed.**
   a. **Client side (Page 130)**
   b. Server Side
   c. Client & Server Side
   d. None of the all

203. **The construction is used to avoid ---------- errors.**
   a. **compilation (Page 157)**
   b. runtime
   c. design time
   d. none of them

204. **Two tests are considered to be equivalent if it is believed that: if one discovers a defect, the other probably will too, and if one does not discover a defect, ----------**
   a. the other probably may
   b. none of them
   c. **the other probably won't either Page 199**
   d. the other probably will

205. **It was lady named ---------- who actually coin the term "bug" for the fiest time.**
   a. Elisay Chistopher
   b. **Admiral Grace Hopper Page 213**
   c. Ana Nicholson
   d. Jane Hopper

206. **Symptoms of logical errors are ----------**
   a. code is misbehaving
   b. Program doesn't crash, but the flow of program takes odd branches through the code.
   c. Results are the opposite
   d. Output looks strange,
   e. **all of the given Page 218**

207. **First-hand accounts of the problem are always useful in Debugging process.**
   a. **True Page 225**
   b. False

208. **architecture elements are further divided into categories which in total are ----------**
   a. 2
   b. **3 Page 122**
   c. 4

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d. 5

209. **Establishing responsibilities for objects includes**
   a. Generalization Relationships
   b. Specialization Relationships
   c. all of the above
   d. **identifying Association relationships Page 101**

210. **Source files can have the extension ----------**
   a. .*c*++
   b. .*C*
   c. .cpp
   d. **all of the given Page 155**

211. **Which of the following is not a characteristic of software?**
   a. **Software is tangible**
   b. A change to a piece of code may implicitly affect the functions of the rest of the code.
   c. Software is configurable.
   d. Software does not wear and tear.

212. **Which of the following is a fact finding method?**
   a. Site visits
   b. **Prototyping**
   c. Study of similar systems
   d. All of given

   Seven common fact-finding methods (Sampling, Research, Observation, Questionnaires, Interviews, Prototyping, and Joint Requirements Planning) are introduced as a means to discover requirements.

213. **The statements given below are associated with system development, testing and maintenance. Identify the correct statement from among them:**
   a. Some of the activities in the implementation phase are building individual system components, writing of programs and development of user interfaces.
   b. The post implementation review is performed to see that the computer system is working.
   c. None of Given
   d. System development phase consists of a development phase and an implementation phase.

214. **Quantitative methods for assessing the quality of proposed architectural designs are readily available.**

a. True
b. False

215. **A decision table should be used_____**
a. to document all conditional statements
b. to guide the development of the project management plan
c. only when building an expert system
d. when a complex set of conditions and actions appears in a component

216. **Test cases should be designed long before testing begins.**
a. **True Page 467**
b. False

217. **Which of the following are characteristics of testable software?**
a. observability
b. simplicity
c. stability
d. **all of the given Page 469**

218. **Comparison testing is typically done to test two competing products as part of customer market analysis prior to product release.**
a. True
b. False

219. **By collecting software metrics and making use of existing software reliability models it is possible to develop meaningful guidelines for determining when software testing is done.**
a. True
b. False

220. **Configuration reviews are not needed if regression testing has been rigorously applied during software integration.**
a. True
b. False

221. **A change becomes ---------- because of close presence of data and fucntions**
a. **Localized Page 81**
b. Private
c. Global
d. Accessible

222. **Requirement engineering mainly deals with the ---------- of the system**
a. **definition phase Page 16**
b. development phase
c. maintenance

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d. none of the above

223. **40-60% of all defects found in software projects can be traced back to poor ----------**

a. **Requirements Page 17**
b. Design
c. Coding
d. Testing

224. **In the N-Tire Architecture the idea is to enhance scalability and ---------- by distributing both the data and the application using multiple server machines.**

a. **Performance Page 131**
b. Efficiency
c. Usability
d. none of the all

225. **MVC pattern was based on the ---------- pattern.**

a. **Observer Page 140**
b. Structural
c. Behavioral
d. None of them

226. **Classes should be declared in individual header files with the file name matching the ---------- name.**

a. **class Page 155**
b. method
c. object
d. none of the all

227. **Unit testing is roughly equivalent to ---------- testing for hardware in which each chip is tested thoroughly after manufacturing**

a. Circuit level
b. **Chip Level Page 207**
c. component level
d. system level

228. **Software should be tested more like hardware, with Built-in self testing: such that each unit can be tested ----------**

a. Freely
b. **independently Page 207**
c. dependtly
d. completely

229. **A system ---------- is the period in which tremendous pressure is on developers end to fix the problem and make the system running again**

a. Uptime

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b. **Downtime Page 214**
c. Currentime
d. Futuretimr

230. **Which one of the given below is not a symptom of memory overrun?**
a. Program crashes quite regularly after a given routine is called, that routine should be examined for a possible overrun condition.
b. If the routine in question does not appear to have any such problem the most likely cause is that another routine, called in the prior sequence, has already trashed variables or memory blocks.
c. Checking the trace log of the called routines leading up to one with the problem will often show up the error.
d. **Compiler warnings. Page 220**

231. **Three tier architecture contains ---------- layers**
a. Presentation
b. Application
c. Database
d. **All of the above Page 131**

232. **Which of the following is a/are tool used in requirement analysis?**
a. Flow Graphs
b. **Data Flow Diagrams**
c. Activity Networks
d. Module Dependency Diagrams

233. **Consider the following comment. "The software which I bought won't run on windows and when it runs I can't use WORD at the same time". Which of the following do you think are violated by the newly bought software?**
a. dependability, interchangeability
b. **platform independence, interoperability**
c. reliability, dependency
d. interoperability, reliability

234. **…… is a diagramming technique used to identify the types of objects in the system and the static relationships that exist among them?**
a. Class Diagram
b. Document flow diagrams
c. Data flow diagrams
d. **Flow charts**

235. **Which of the following interaction style best suit the design of an interface for visually impaired users?**
a. **direct manipulation Page 469**
b. direct manipulation

c. natural language
d. command line

236. **Which of the following testing involve purely black box testing?**
a. unit testing, beta testing
b. acceptance testing, interface testing
c. beta testing, acceptance testing
d. **integration testing, interface testing**

237. **What are the three generic phases of software engineering?**
a. **definition, development, support Page 96**
b. what, how, where
c. programming, debugging, maintenance
d. analysis, design, testing

238. **In the context of requirements analysis, partitioning results in the elaboration of data, function, or behavior.**
a. True
b. False

239. **In refining the DFD during transaction mapping it is unnecessary to create a PSPEC since only the CSPEC is relevant to this type of architectural style.**
a. True
b. False

240. **In transaction mapping the first level factoring results in the_____**
a. creation of a CFD
b. derivation of the control hierarchy
c. distribution of worker modules
d. refinement of the module view

241. **Variable names must be in mixed case starting with upper case.**
a. True
b. **False Page 150**

242. **Class variables should be declared public. This concept violates which of the following.**
a. Information hiding
b. Encapsulation
c. **Information hiding and Encapsulation Page 158**
d. None of given

243. **Real-time applications add a new and potentially difficult element to the testing mix**
a. performance
b. reliability

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c. security
d. **time Page 498**

244. **Top-down integration testing has as it's major advantage(s) that**
a. low level modules never need testing
b. major decision points are tested early
c. no stubs need to be written
d. none of the given

245. **is Authorized, assess Performance, get item for UPC, get cashier for number, are the examples of**
a. **Services Page 102**
b. Attributes
c. Links
d. None of the above

246. **the *development* view describes the ----------organization of the software in its development environment,**
a. **Static Page 122**
b. Dynamic
c. Still
d. Static & Dynamic

247. **Hungarian Notation was first discussed by Charles Simonyi of----------**
a. **Microsoft. Page 149**
b. Oracle
c. Apple Macintosh
d. None of the all

248. **The language does not specify whether ---------- is signed or unsigned.**
a. **Char Page 181**
b. integer
c. double
d. constant

249. **A number of invisible execution paths can exist in simple code in a language that allows ----------**
a. **Exceptions Page 185**
b. defects
c. errors
d. all of them

250. **Bugs Fixing is done by which of the teams in Software Development lifecycle?**
a. **Development Team Page 196**
b. Testing Team

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c. Analysis & Design Team
d. Process Team

251. **Software Bugs have multiple names, Which one of the below is not the name of Software Bugs ----------**
a. Bugs
b. Defects
c. Errors
d. **Mistakes Page 213**

252. **In the debugging process a stack trace is a very useful tool.**
a. **True Page 226**
b. False

253. **Stakeholders are different people who would be interested in the ----------**
a. **Software Page 24**
b. System
c. Product
d. All of the given

254. **Which of the following is an/are advantage(s) of object oriented analysis & design over structured system analysis & design?**
a. Ease of modeling real world events
b. Reusability
c. Maintainability
d. All of given

255. **The two main documents produced during this phase are Requirement Statement and Requirement Specification. They are also called Requirement Definition and ----------**
a. **Functional Specification Page 25**
b. Mathematical specification
c. System Specification
d. None of the given

256. **A ---------- is a code that explains itself without the need of comments and extraneous documentation**
a. **Self-documenting code Page 147**
b. Self-telling Code
c. Self-Documenting Design
d. None of the above

257. **Bugs that won't "stand still" (almost random) are the ---------- to deal with.**
a. least difficult
b. **most difficult Page 226**

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c. very easy
d. easy

258. **Association is a type of relation**
a. **Weak OOP , 49**
b. Stronger
c. Normal
d. None of above

259. **Bugs that won't (almost random) are the ---------- to deal with.**
a. **stand still Page 226**
b. Dynamic
c. Static
d. None of above

260. **CRUD include following:**
a. Creat
b. Update
c. Read
d. **All of above Page 53**

261. **Project ---------- defines the concept and range of the proposed solution, and limitations identify certain capabilities that the product will not include**
a. **Scope Page 30**
b. Agreement
c. Plan
d. None of the given

262. **The cyclomatic complexity metric provides the designer with information regarding the number of**
a. Cycles in the program
b. Errors in the program
c. **Independent logic paths in the program**
d. Statements in the program

263. **Which of the followings is not a testing type?**
a. Subsystem testing
b. Alpha testing
c. Beta Testing
d. **Gamma Testing Page 198**

264. **Which of the items listed below is not one of the software engineering layers?**
a. Process
b. **Manufacturing**
c. Methods

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d. Tools

265. **Which piece of code is self-documented?**
   a. if (x==0) // this is the case when we are allocating a new number
   b. if (AllocFlag == 0)
   c. **If (AllocFlag == NEW_NUMBER) Page 148**
   d. None of the given

266. **UML (unified modeling language) analysis modeling focuses on the _____**
   a. Behavioral model and environment model.
   b. **B**ehavioral model and implementation model.
   c. user model and environmental model
   d. **user model and structural model Page 604**

267. **What makes requirements elicitation difficult?**
   a. bounding scope
   b. understanding user needs
   c. requirements volatility
   d. **all of the above Page 285**

268. **In the architecture trade-off analysis method the architectural style should be described using the**
   a. module view
   b. process view
   c. **all of the given Page 403**
   d. data flow view

269. **Which of these are objectives for software testing?**
   a. determine the productivity of programmers
   b. eliminate the need for future program maintenance
   c. eliminate every error prior to release
   d. **uncover software errors Page 467**

270. **In software quality assurance work there is no difference between software verification and software validation**
   a. True
   b. False

271. **Top-down integration testing has as it's major advantage(s) that**
   a. low level modules never need testing
   b. major decision points are tested early
   c. no stubs need to be written
   d. none of the given

272. **Rooko mut jane do depicts one of risks associated with inadequate requriements gathering, the name of that risk is ----------**
   a. **Ambiguous requirements Page 20**
   b. Creeping Requirements
   c. Minimal Specification
   d. None of the above

273. **Interaction Diagrams depict the ---------- Behavior of the system**
   a. Static
   b. **Dynamic Page 106**
   c. Active
   d. None of the above

274. **In Sequence Diagrams events are organized in a----------time line**
   a. Vertical
   b. **Horizontal Page 106**
   c. Vertical & Horizontal
   d. None of the above

275. **object-oriented framework that is a set of cooperative classes that make up reusable _____of a system**
   a. Code
   b. **Design Page 138**
   c. Analysis
   d. none of the all

276. **In order to make a code more portable, Instead of using vendor specific language extensions, use ---------- as much as possible**
   a. **STL Page 179**
   b. ANSI
   c. ISO
   d. CMMI

277. **Performance & Usability are examples of ---------- requirements**
   a. Business
   b. Functioanl
   c. **Non-Functioanl Page 39**
   d. User

278. **According to a survey, when a software application is in the maintenance phase, --------- of its lifecycle cost is attributed towards the defects which are found in the software application after installation**
   a. **20 % Page 214)**
   b. 25 %

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c. 30%
d. 35%

279. **Bugs that won't "----------" (almost random) are the most difficult to deal with.**
a. **stand still** **Page 226**
b. dynamic
c. running
d. stand tall

280. **If you are initializing a variable at the time of declaration, do not declare another _____ in the same statement.**
a. **variable** **Page 176**
b. object
c. class
d. None of given

281. **During Requirement Engineering process which of the following is/are document(s) used for fact finding?**
a. Company's employee list
b. Samples of the company databases
c. The company's mission statement and plan
d. **All of given**

282. **Object-oriented domain analysis is concerned with the identification and specification of reusable capabilities within an application domain.**
a. True
b. False

283. Fourth generation techniques
a. Allow software to be developed without any testing.
b. Eliminate the need for costly requirements gathering activities.
c. Can reduce the time required to develop software. **Page 73**
d. Are best used by non-programmers to build small systems

284. Data design actually begins during the creation of the analysis model, not the architectural model.
a. **True** **page397**
b. False

285. The states shown in a state transition diagram do not necessarily correspond to the processes shown in a control flow diagram for the same system.
a. **True** **page397**
b. False

286. The criteria used to assess the quality of an architectural design should be based on system
a. **accessibility and reliability** **page404**

b. data and control
c. functionality
d. implementation details

287. **A useful technique for evaluating the overall complexity of a proposed architecture is to look at the component**
a. number and size of components
b. **flow dependencies and sharing dependencies Page 406**
c. size and cost
d. none of the given

288. The three basic principles that guide maintainability are: simplicity, clarity, and _____.
a. **Generality Page 146**
b. Reliability
c. All of the given choices

289. **In order to make a program self-documented a number of attributes required. Which one is the attribute/s of self-documented program**
a. **All of the given choices Page 147**
b. Size of each function
c. Choice of variable

290. Floating point constants should always be written with decimal point and at least
a. **one decimal Page 161**
b. two decimal
c. three decimal

291. **The code becomes self-explanatory with the help of proper use of parentheses. Select the right one.**
a. leapYear = year % 4 == 0 && year % 100 != 0 || year % 400 == 0;
b. leapYear = ((year % 4 == 0) && (year % 100 != 0) || ((year % 400 == 0));
c. **leapYear = ((year % 4 == 0) && (year % 100 != 0)) || (year % 400 == 0); Page 164**
d. leapYear = (year % 4 == 0) && (year % 100 != 0) || (year % 400 == 0);

292. **Graph-based testing methods can only be used for object-oriented system**
a. True
b. False

293. **A class is a ---------- of objects**
a. Template page85
b. Stereotype
c. Collection
d. None of above

294. A public Interface provides a way for with other Classes.
   a. Communication
   b. Accessibility
   c. Reaching
   d. All of the above

295. **Software architecture defines the high level structure of the software by putting together a number of architectural ---------- in an organized fashion.**
   a. **Elements Page 122**
   b. **Parts**
   c. **Components**
   d. **None of all**

296. **Patterns are devices that allow programs to share knowledge about their ----------.**
   a. **Design Page 137**
   b. Code
   c. Analysis
   d. None of all

297. **Type conversions must always be done ----------**
   a. **Explicitly Page 158**
   b. Implicitly
   c. Simultaneously
   d. None of them

298. **Types that are ----------to one file only can be declared inside that file**
   a. Local page158
   b. Global
   c. Private
   d. General

299. **The use of do while loops should be ----------**
   a. Avoided page159
   b. Encouraged
   c. Practiced
   d. None of them

300. **One of the causes of the portability issues is the order of ---------- varies from one implementation to other.**
   a. **Evaluation page181**
   b. **Numbers**
   c. **Variables**
   d. Symbols

301. **The goal of testing is to expose ---------- defects in a software system before it is put to use.**

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a. **Latent** **page193**
b. **Already present**
c. **Current**
d. **Runtime**

302. In the N-tier architecture, the idea is to enhance scalability and ------ by distributing both the data and the application using multiple server machines.
a. Usability
b. Performance **page # 134**
c. Interoperability
d. None

303. In case of a file servers, client requests selected records from a ….. and the server transmits records to client over the network.
a. Local memory
b. Network
c. Database
d. File **page #132**

304. Alpha testing is:
a. Testing individual components independent of other components
b. Testing a collection of dependent components
c. Validation against user expectations
d. **Acceptance testing for customized projects, in-house testing for products page#201**

305. This code uses the % operator to set the rear pointer to 0 once it has reached
a. Min-Size
b. Max-Size **Page 169**
c. Both
d. None

306. Which of these represents the Krutchen's 4+1 architectural view model?
a. Logical view, Process view, Physical view, Development view, Use case view
b. Logical view, Dynamic view, Physical view, Development view, User case view
c. Logical view, Process view, Physical view, Development view, Sequence view
d. Dynamic view, Process view, Physical view, Development view, Use case view

307. Vertical participation divides the architecture application from a making perspective.
a. Decision
b. Design
c. Conclusion
d. Move

308. Anti- Patterns is another concept that corresponds to common in analysis and design.
a. Mistake
b. Issues
c. Problems
d. All of the given

309. block of error-sensitive code with exception handling is called
a. Trying to execute a block

310. Which function throw for several reasons but all that matters for this function is whethere

a. Title ()

311. The best reason for using Independent software test teams is that
a. Software developers do not need to do any testing
b. Strangers will test the software mercilessly
c. Testers do not get involved with the project until testing begins
d. The conflicts of interest between developers and testers is reduced

312. _____ is intended to define a many to many relationship between objects so that when
a. One object changes state all its dependents are notified and updated automatically.
b. Observer Pattern
c. Facade Pattern
d. Singleton Pattern
e. Joint Pattern

313. Secondary private classes can be declared as _____ and reside in the file of the class
a. Asynchronous Classes
b. Inner classes PAGE158
c. Outer classes
d. Synchronous classes

314. The construction should appear at the _____ of the header file.
a. Top PAGE 160
b. Bottom
c. Left
d. Right

315. Modularity is a tool that can help us in _____ the size of individual functions.
a. Stabilizing
b. Reducing PAGE 173
c. Increasing
d. Strengthening

316. Abstraction and encapsulation are two important tools that can help in managing and mastering the _____ of a program.
a. Usability
b. Complexity Page173
c. Understandability
d. Reliability

317. Which of the following is/are NOT one of the umbrella activities?
a. Requirement analysis
b. Architecture design
c. Test case development
d. All of the given options Page 14

318. Which one is NOT a type of messages which Sequence Diagrams Depict _____
a. Asynchronous
b. Synchronous
c. Create
d. Update Page 111

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319. Identifying Whole-Part structures (Aggregations) means, what are my ----------
   a. Components PAGE 98
   b. Structures
   c. Modules
   d. Interaction Protocols
320. The design process for identifying the sub-systems making up a system and the Framework for sub-system control and communication is:
   a. Architectural Design PAGE 118
   b. Interface Design
   c. Component Design
   d. Data Design
321. A complex System evolves from a
   a. Smaller system
   b. Simpler system (GOOG)
   c. Bigger system
   d. Medium system
322. A context diagram _____
   a. Describes the context dependencies of a system
   b. Is a DFD which gives an overview of the system (Conceptual)
   c. Is a detailed description of a system
   d. Is not used in drawing a detailed DFD
323. When measure of independence of a module or component is low to the other, How would changes in one component have effect on other component?
   a. No
   b. Low
   c. High (Conceptual)
   d. Equal
324. What is meant by the term 'software crisis'?
   a. Ability to deliver software on time with good quality
   b. A situation in which experienced developers leave the company
   c. Inability of new software to inter-operate with existing software (Conceptual)
   d. A situation in which large scale software projects took more time and budget than was planned
325. Equivalence _____ help you in designing test cases to test the system effectively and efficiently.
   a. Functions
   b. Classes PAGE 202
   c. Objects
   d. Interfaces
326. _____ requirements are often called product features.
   a. Functional
   b. Business
   c. User
   d. Non-functional
327. Many compilers limit the maximum number of bits in the bit field to the size of a(n)

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_____.

a. Integer PAGE186

b. Float

c. Character

d. Double

328. In Java, ">>" is used for _____ shift and ">>>" for _____ shift.

a. Arithmetic, Logical PAGE 184

b. Mathematical, Logical

c. Incremental, Arithmetic

d. Logical, Arithmetic

329. _____ is only a reference for defining protocols and designing and implementing systems developed by different parties.

a. OSI model

b. Reference architecture PAGE 137

c. Layered architecture

d. N-tier architecture

330. Testing activities require destructive instincts in _____ for the purpose of breaking system to discover loopholes into its functionality.

a. Bug Fixers

b. Tester PAGE 198

c. Developer

d. Requirement Engineer

331. In presence of _____ bug in a program, the results are the opposite of what is expected.

a. Memory Leak

b. Memory Over-runs

c. Syntax Error

d. Logical Error Page 221

332. Which of the following is a non-functional requirement of a website which sells songs?

a. A catalogue of the stock needs to be available for the users to choose from

b. Customer information should be retained to allow future transactions easier

c. Users should be able to choose from a set of different languages for the interface

d. Time taken to download songs in the catalogue should not irritate users (CONCEPTUAL)

333. The state transition diagram _____

a. Depicts relationships between data objects

b. Depicts functions that transform the data flow

c. Indicates how data are transformed by the system (CONCEPTUAL)

d. Indicates system reactions to external events

334. Comments are not syntax checked

a. TRUE PAGE 165

b. FALSE

335. Modularity is a tool that can help us in increasing the size of individual functions, making them less readable.

a. True

b. False PAGE 173

336. Modules with high cohesion and low coupling can be treated and analysed as
   a. White boxes
   b. black boxes Page 75
   c. grey boxes
   d. none of these
337. While establishing the services for an object, the following fundamental questions should be asked:
   a. Why does the system need this object anyway?
   b. What useful questions can it answer?
   c. What useful action can it perform?
   d. All of the given options.
338. ----------is a role that each actor plays in the system under consideration.
   a. An act
   b. A participant PAGE 96
   c. A function
   d. None of the given
339. Any Engineering approach must be founded on organizational commitment to ---------
   a. Cost
   b. Scheduling
   c. Quality Page 115
   d. Performance
340. Return values in synchronous massages are:
   a. Compulsory
   b. May not used when response is obvious
   c. Not used at all
   d. Represented by solid lines
341. According to Caper Jhones analysis of project activities, coding only has ------- affect part in system development.
   a. 13-14%
   b. 36-40%
   c. 50-60%
   d. 70-80%
342. Following are some statements associated with data flow diagrams. Identify the correct statement from among them.
   a. DFDs are used to model complex interfaces.
   b. DFDs are used to represent only functional processing, data stores and data movements between functions.
   c. DFDs depict only processes which can be decomposed.
   d. DFDs do not show external data sources and external data sinks.
343. A project is considered successful if:
   a. The system was delivered in time and within budget.
   b. The system meets at least some of the customer's requirements.
   c. The system development process has a maximum impact on the ongoing business process.
   d. Minimum time was spent for requirement gathering and designing.

344. A memory leak bug is one in which memory is somehow allocated from either the operating system or an ---------- "pool", but never deallocated when the memory is finished being used
   a. Mixed memory
   b. External memory
   c. Internal Memory Page 219
   d. Mutually exclusive memory

345. Control flow diagrams are:
   a. Needed to model event driven systems
   b. Required for all systems
   c. Used in place of data flow diagrams
   d. Useful for modeling user interfaces

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FINALTERM EXAMINATION
Spring 2010
CS504- Software Engineering - I (Session - 2)

Question No: 1 (Marks: 1) - Please choose one
The first step in any OOA process model is to_____

- ► build an object-relationship model
- ► define collaborations between objects
- ► **elicit customer requirements** **(A PRACTITIONER'S APPROACH Page603)**
- ► select a representation language

Question No: 2 (Marks: 1) - Please choose one
A necessary supplement to transform or transaction mapping needed to create a complete architectural design is _____.

- ► entity relationship diagrams
- ► the data dictionary
- ► **processing narratives for each module** **(A PRACTITIONER'S APPROACH Page422)**
- ► test cases for each module

Question No: 3 (Marks: 1) - Please choose one
What types of errors are missed by black-box testing and can be uncovered by white-box testing?

- ► behavioral errors
- ► **subtle logic errors** **Click here for detail**
- ► performance errors
- ► input error

Question No: 4 (Marks: 1) - Please choose one
Which test criteria should be applied in a phase of testing?

- ► functional validity
- ► interface integrity
- ► correctness
- ► **all of the given** **(A PRACTITIONER'S APPROACH Page522)**

Question No: 5 (Marks: 1) - Please choose one
The hardest single part of building a software system is deciding precisely ---------- to build.

- ► **what (Page 17)**
- ► How
- ► When
- ► Why

Question No: 6 (Marks: 1) - Please choose one
In sequence Diagrams objects are organized in a ----------line

- ► **horizontal (Page 106)**
- ► vertical
- ► horizontal and vertical
- ► non of the above

Question No: 7 (Marks: 1) - Please choose one
Every view is potentially Hierarchical in this regard which of the view are Hierarchical ?

- ► Functioanl View
- ► Development View
- ► *Concurrency view*
- ► **All of given (Page 125)**

Question No: 8 (Marks: 1) - Please choose one
Like analysis models, many different kinds of ----------models are developed

- ► **Architectural (Page 126)**
- ► System
- ► Design
- ► Logical

Question No: 9 (Marks: 1) - Please choose one
Idea behind zero install architecture is to develop a system where no installation on the ---------- is needed.

- ► **Client side (Page 130)**
- ► Server Side
- ► Client & Server Side
- ► Non of the all

Question No: 10 (Marks: 1) - Please choose one
The construction is used to avoid ---------- errors.

- ► **compilation (Page 157)**
- ► runtime
- ► design time
- ► non of them

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Question No: 11 (Marks: 1) - Please choose one
All comments should be written in----------

- **English (Page 162)**
- French
- C++
- JAVA

Question No: 12 (Marks: 1) - Please choose one
One of the Guidelines to avoid common mistakes is to never use --------- except for declaration

- **, (Page 176)**
- ;
- =
- ||

Question No: 13 (Marks: 1) - Please choose one
The order in which bytes of one word are stored is ---------- dependent

- **Hardware (Page 181)**
- Software
- language
- system

Question No: 14 (Marks: 1) - Please choose one
Verification and validation are the processes in which we check a product against its --------- and the ---------- of the users who will be using it

- **specifications, expectations (Page 192)**
- statements, expectations
- specifications , arguments
- statements, arguments

Question No: 15 (Marks: 1) - Please choose one
Two tests are considered to be equivalent if it is believed that: if one discovers a defect, the other probably will too, and if one does not discover a defect, ----------.

- the other probably may
- non of them
- **the other probably won't either (Page 199)**
- the other probably will

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Question No: 16 (Marks: 1) - Please choose one
It was lady named ---------- who actually coin the term “bug" for the fiest time.

- ► Elisay Chistopher
- ► **Admiral Grace Hopper (Page 213)**
- ► Ana Nicholson
- ► Jane Hopper

Question No: 17 (Marks: 1) - Please choose one
The first “bug” was actually a moth, which flew through an open window and into one of the Mark ----------’s relays.

- ► **II (Page 213)**
- ► I
- ► III
- ► IV

Question No: 18 (Marks: 1) - Please choose one
Symptoms of logical errors are ----------

- ► code is misbehaving
- ► The program doesn't crash, but the flow of program takes odd branches through the code.
- ► Results are the opposite
- ► Output looks strange,
- ► **all of the given (Page 218)**

Question No: 19 (Marks: 1) - Please choose one
First hand accounts of the problem are always useful in Debugging process.

- ► **True (Page 225)**
- ► False

Question No: 20 (Marks: 1) - Please choose one
Software architecture elements are further divided into categories which in total are ----------

- ► 2
- ► **3 (Page 122)**
- ► 4
- ► 5

Question No: 21 (Marks: 1) - Please choose one
Establishing responsibilities for objects includes

- ► Generalization Relationships
- ► Specialization Relationships
- ► all of the above
- ► **identifying Association relationships (Page 101)**

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Question No: 22 (Marks: 1) - Please choose one
Source files can have the extension ----------

- ► *c++* .
- ► .*C*
- ► .cpp
- ► **all of the given (Page 155)**

Question No: 23 (Marks: 1) - Please choose one
Be very careful when you use functions with side effects – functions that change the values of the ----------

- ► **parameters (Page 176)**
- ► variables
- ► constants
- ► None of given

Question No: 24 (Marks: 1) - Please choose one
Which of the following is not a characteristic of software?

- ► **Software is tangible**
- ► A change to a piece of code may implicitly affect the functions of the rest of the code.
- ► Software is configurable.
- ► Software does not wear and tear.

Question No: 25 (Marks: 1) - Please choose one
Which of the following is a fact finding method?

- ► Site visits
- ► **Prototyping**
- ► Study of similar systems
- ► All of given

Seven common fact-finding methods (Sampling, Research, Observation, Questionnaires, Interviews, Prototyping, Joint Requirements Planning) are introduced as a means to discover requirements.

Question No: 26 (Marks: 1) - Please choose one
The statements given below are associated with system development, testing and maintenance. Identify the correct statement from among them:

► Some of the activities in the implementation phase are building individual system components, writing of programs and development of user interfaces.
► The post implementation review is performed to see that the computer system is working.
► None of Given
► System development phase consists of a development phase and an implementation phase.

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FINALTERM EXAMINATION
Spring 2010

Question No: 1 (Marks: 1) - Please choose one
Quantitative methods for assessing the quality of proposed architectural designs are readily available.

- ► True
- ► **False** http://highered.mcgraw-hill.com/sites/0072853182/student_view0/chapter10/multiple_choice_quiz.html

Question No: 2 (Marks: 1) - Please choose one
A decision table should be used_____

- ► to document all conditional statements
- ► to guide the development of the project management plan
- ► only when building an expert system
- ► **when a complex set of conditions and actions appears in a component** **Click here for detail**

Question No: 3 (Marks: 1) - Please choose one
Test cases should be designed long before testing begins.

- ► **True** **(A PRACTITIONER'S APPROACH Page 467)**
- ► False

Question No: 4 (Marks: 1) - Please choose one
Which of the following are characteristics of testable software?

- ► observability
- ► simplicity
- ► stability
- ► **all of the given** **(A PRACTITIONER'S APPROACH Page 469)**

Question No: 5 (Marks: 1) - Please choose one
Comparison testing is typically done to test two competing products as part of customer market analysis prior to product release.

- ► True
- ► **False** **Click here for detail**

Question No: 6 (Marks: 1) - Please choose one
By collecting software metrics and making use of existing software reliability models it is possible to develop meaningful guidelines for determining when software testing is done.

- ► **True** **Click here for Detail**
- ► False

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Question No: 7 (Marks: 1) - Please choose one
Configuration reviews are not needed if regression testing has been rigorously applied during software integration.

- ► True
- ► **False** **Click here for detail**

Question No: 8 (Marks: 1) - Please choose one
A change becomes ---------- because of close presence of data and fucntions

- ► **Localized (Page 81)**
- ► Private
- ► Global
- ► Accessible

Question No: 9 (Marks: 1) - Please choose one
Requirement engineering mainly deals with the ---------- of the system

- ► **definition phase (Page 16)**
- ► development phase
- ► maintenance
- ► non of the above

Question No: 10 (Marks: 1) - Please choose one
40-60% of all defects found in software projects can be traced back to poor ----------

- ► **Requirements (Page 17)**
- ► Design
- ► Coding
- ► Testing

Question No: 11 (Marks: 1) - Please choose one
In the N-Tire Architecture the idea is to enhance scalability and ---------- by distributing both the data and the application using multiple server machines.

- ► **Performance (Page 131)**
- ► Efficiency
- ► Usability
- ► non of the all

Question No: 12 (Marks: 1) - Please choose one
MVC pattern was based on the ---------- pattern.

- ► **Observer (Page 140)**
- ► Structural
- ► Behavioral
- ► Non of them

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Question No: 13 (Marks: 1) - Please choose one
Classes should be declared in individual header files with the file name matching the ---------- name.

► **class (Page 155)**
► method
► object
► non of the all

Question No: 14 (Marks: 1) - Please choose one
Unit testing is roughly equivalent to ---------- testing for hardware in which each chip is tested thoroughly after manufacturing

► Circuit level
► **Chip Level (Page 207)**
► component level
► system level

Question No: 15 (Marks: 1) - Please choose one
Software should be tested more like hardware, with Built-in self testing: such that each unit can be tested ----------

► freely
► **independently (Page 207)**
► dependtly
► completely

Question No: 16 (Marks: 1) - Please choose one
A system ---------- is the period in which tremendous pressure is on developers end to fix the problem and make the system running again

► Uptime
► **Downtime (Page 214)**
► Currentime
► futuretimr

Question No: 17 (Marks: 1) - Please choose one
Which one of the given below is not a symptom of memory overrun?

► Program crashes quite regularly after a given routine is called, that routine should be examined for a possible overrun condition.
► If the routine in question does not appear to have any such problem the most likely cause is that another routine, called in the prior sequence, has already
trashed variables or memory blocks.
► Checking the trace log of the called routines leading up to one with the problem will often show up the error.
► **Compiler warnings. (Page 220)**

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Question No: 18 (Marks: 1) - Please choose one
Software architecture elements are further divided into categories which in total are ----------

- ► 2
- ► **3 (Page 122) rep**
- ► 4
- ► 5

Question No: 19 (Marks: 1) - Please choose one
Three tier architecture contains ---------- layers

- ► Presentation
- ► Application
- ► Database
- ► **All of the above (Page 131)**

Question No: 20 (Marks: 1) - Please choose one
1. Variables should be initialized where they are declared and they should be declared in the ---------- scope possible.

- ► **Smallest (Page 158)**
- ► largest
- ► medium
- ► None of the given

Question No: 21 (Marks: 1) - Please choose one
---------- are two important tools that can help in managing and mastering the complexity of a program.

- ► **abstraction and encapsulation (Page 170)**
- ► abstraction and Inheritence
- ► abstraction and Polymarhphism
- ► None of given

Question No: 22 (Marks: 1) - Please choose one
Which of the following is a/are tool used in requirement analysis?

- ► Flow Graphs
- ► **Data Flow Diagrams (not sure)**
- ► Activity Networks
- ► Module Dependency Diagrams

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Question No: 23 (Marks: 1) - Please choose one
Consider the following comment. "The software which I bought won't run on windows and when it runs I can't use WORD at the same time". Which of the following do you think are violated by the newly bought software?

- ► dependability, interchangeability
- ► **platform independence, interoperability**
- ► reliability, dependency
- ► interoperability, reliability

Question No: 24 (Marks: 1) - Please choose one
…… is a diagramming technique used to identify the types of objects in the system and the static relationships that exist among them?

- ► Class Diagram
- ► Document flow diagrams
- ► Data flow diagrams
- ► **Flow charts**

Question No: 25 (Marks: 1) - Please choose one
Which of the following interaction style best suit the design of an interface for visually impaired users?

- ► **direct manipulation** **(A PRACTITIONER'S APPROACH Page 469)**
- ► direct manipulation
- ► natural language
- ► command line

Question No: 26 (Marks: 1) - Please choose one
Which of the following testing involve purely black box testing?

- ► unit testing, beta testing
- ► acceptance testing, interface testing
- ► beta testing, acceptance testing
- ► **integration testing, interface testing**

FINALTERM EXAMINATION
Spring 2010

Question No: 1 (Marks: 1) - Please choose one
What are the three generic phases of software engineering?

- ► **definition, development, support** **(A PRACTITIONER'S APPROACH Page 96)**
- ► what, how, where
- ► programming, debugging, maintenance
- ► analysis, design, testing

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Question No: 2 (Marks: 1) - Please choose one
In the context of requirements analysis, partitioning results in the elaboration of data, function, or behavior.

► **True** **Click here for detail**
► False

Question No: 3 (Marks: 1) - Please choose one
Quantitative methods for assessing the quality of proposed architectural designs are readily available.
► True
► **False** **Click here for detail**

Question No: 4 (Marks: 1) - Please choose one
In refining the DFD during transaction mapping it is unnecessary to create a PSPEC since only the CSPEC is relevant to this type of architectural style.
► True
► **False** **click here for detail**

Question No: 5 (Marks: 1) - Please choose one
In transaction mapping the first level factoring results in the_____

► creation of a CFD
► **derivation of the control hierarchy** **Click here for detail**
► distribution of worker modules
► refinement of the module view

Question No: 6 (Marks: 1) - Please choose one
Variable names must be in mixed case starting with upper case.
► True
► **False (Page 150)**

Question No: 7 (Marks: 1) - Please choose one
Class variables should be declared public. This concept violates which of the following.
► Information hiding
► Encapsulation
► **Information hiding and Encapsulation (Page 158)**
► None of given

Question No: 8 (Marks: 1) - Please choose one
Real-time applications add a new and potentially difficult element to the testing mix

► performance
► reliability
► security
► **time** **(Page 498)**

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Question No: 9 (Marks: 1) - Please choose one
Top-down integration testing has as it's major advantage(s) that

- ► low level modules never need testing
- ► **major decision points are tested early** **Click here for detail**
- ► no stubs need to be written
- ► none of the given

Question No: 10 (Marks: 1) - Please choose one
Configuration reviews are not needed if regression testing has been rigorously applied during software integration.

- ► True
- ► **False** **Click here for detail**

Question No: 11 (Marks: 1) - Please choose one
isAuthorized, assess Performance,get item for UPC, get cashier for number, are the examples of

- ► **Services (Page 102)**
- ► Attributes
- ► Links
- ► Non of the above

Question No: 12 (Marks: 1) - Please choose one
the *development* view describes the ----------organization of the software in its development environment,

- ► **Static (Page 122)**
- ► Dynamic
- ► Still
- ► Static & Dynamic

Question No: 13 (Marks: 1) - Please choose one
---------- being the first pure Object Oriented language in which observer pattern was used in implementing its Model View Controller

- ► **Smalltalk (Page 140)**
- ► PASCAL
- ► JAVA
- ► C++

Question No: 14 (Marks: 1) - Please choose one
Hungarian Notation was first discussed by Charles Simonyi of----------

- ► **Microsoft. (Page 149)**
- ► Oracle
- ► Apple Macintosh
- ► Non of the all

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Question No: 15 (Marks: 1) - Please choose one
The language does not specify whether ---------- is signed or unsigned.

- ► **char (Page 181)**
- ► integer
- ► double
- ► constant

Question No: 16 (Marks: 1) - Please choose one
A number of invisible execution paths can exist in simple code in a language that allows ----------

- ► **exceptions (Page 185)**
- ► defects
- ► errors
- ► all of them

Question No: 17 (Marks: 1) - Please choose one
Bugs Fixing is done by which of the teams in Software Development lifecycle?

- ► **Development Team (Page 196)**
- ► Testing Team
- ► Analysis & Design Team
- ► Process Team

Question No: 18 (Marks: 1) - Please choose one
Software Bugs have multiple names, Which one of the below is not the name of Software Bugs ----------
-

- ► Bugs
- ► Defects
- ► Errors
- ► **Mistakes (Page 213)**

Question No: 19 (Marks: 1) - Please choose one
Symptoms of logical errors are ----------

- ► code is misbehaving
- ► The program doesn't crash, but akes odd branches through the code.
- ► Results are the opposite
- ► Output looks strange,
- ► **all of the given (Page 218)**

Question No: 20 (Marks: 1) - Please choose one
First hand accounts of the problem are always useful in Debugging process.

- ► **True (Page 225)**
- ► False

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Question No: 21 (Marks: 1) - Please choose one
In the debugging process a stack trace is a very useful tool.

- ► **True (Page 226)**
- ► False

Question No: 22 (Marks: 1) - Please choose one
Establishing responsibilities for objects includes

- ► Generalization Relationships
- ► Specialization Relationships
- ► all of the above
- ► **identifying Association relationships (Page 101)**

Question No: 23 (Marks: 1) - Please choose one
Stakeholders are different people who would be interested in the ----------

- ► **Software (Page 24)**
- ► System
- ► Product
- ► All of the given

Question No: 24 (Marks: 1) - Please choose one
A ---------- is a code that explains itself without the need of comments and extraneous documentation

- ► **Self documenting code (Page 147)**
- ► Self telling Code
- ► Self Documenting Design
- ► None of the given

Question No: 25 (Marks: 1) - Please choose one
1. Variables should be initialized where they are declared and they should be declared in the ---------- scope possible.

- ► **Smallest (Page 158)**
- ► largest
- ► medium
- ► None of the given

Question No: 26 (Marks: 1) - Please choose one
Which of the following is an/are advantage(s) of object oriented analysis & design over structured system analysis & design?

- ► Ease of modeling real world events
- ► Reusability
- ► Maintainability
- ► **All of given** **Click here for detail**

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FINAL TERM EXAMINATION SPRING 2010

Question No: 1 (Marks: 1) - Please choose one
The two main documents produced during this phase are Requirement Statement and Requirement Specification. They are also called Requirement Definition and ----------

- ► **Functional Specification (Page 25)**
- ► Mathematical specification
- ► System Specification
- ► None of the given

Question No: 2 (Marks: 1) - Please choose one
A decision table should be used_____

- ► To document all conditional statements
- ► To guide the development of the project management plan
- ► Only when building an expert system
- ► **when a complex set of conditions and actions appears in a component** **Click here for detail**

Question No: 3 (Marks: 1) - Please choose one
Which of the following is a/are tool used in requirement analysis?

- ► Flow Graphs
- ► **Data Flow Diagrams (not sure)**
- ► Activity Networks
- ► Module Dependency Diagrams

Question No: 4 (Marks: 1) - Please choose one
MVC stands for ----------

- ►**Model View Controller (Page 140)**
- ►Modern View Center
- ►Model View Center
- ►Modern View Controller

Question No: 5 (Marks: 1) - Please choose one
A ---------- is a code that explains itself without the need of comments and extraneous documentation

- ►**Self documenting code (Page 147)**
- ►Self telling Code
- ►Self Documenting Design
- ►Non of the above

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Question No: 6 (Marks: 1) - Please choose one

Bugs that won't "stand still" (almost random) are the ---------- to deal with.

►least difficult

►**most difficult (Page 226)**

►very easy

►easy

Question No: 7 (Marks: 1) - Please choose one

In refining the DFD during transaction mapping it is unnecessary to create a PSPEC since only the CSPEC is relevant to this type of architectural style.

► True

► **False**

Question No: 8 (Marks: 1) - Please choose one

Association is a type _____of relation

►**Weak (OOP , 49)**

►Stronger

►Normal

►None of above

Question No: 9 (Marks: 1) - Please choose one

Variable names must be in mixed case starting with upper case.

► True

► **False (Page 150)**

Question No: 10 (Marks: 1) - Please choose one

Bugs that won't (almost random) are the ---------- to deal with.

►**stand still (Page 226)**

►Dynamic

►Static

►None of above

Question No: 11 (Marks: 1) - Please choose one

CRUD include following:-

►Creat

►Update

►Read

►**All of above (Page 53)**

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Question No: 12 (Marks: 1) - Please choose one
Bugs Finding is done by which of the teams in Software Development lifecycle?

- ► Development Team
- ► **Testing Team (Page 196)**
- ► Analysis & Design Team
- ► Process Team

Question No: 13 (Marks: 1) - Please choose one
Software Bugs have multiple names, which one of the below is not the name of Software Bugs ----------

- ► Bugs
- ► Defects
- ► Errors
- ► **Mistakes (Page 213)**

Question No: 14 (Marks: 1) - Please choose one
In the debugging process a stack trace is a very useful tool.

- ► **True (Page 226)**
- ► False

Question No: 15 (Marks: 1) - Please choose one
A ---------- is a code that explains itself without the need of comments and extraneous documentation

- ► **Self documenting code (Page 147)**
- ► Self telling Code
- ► Self Documenting Design
- ► None of the given

Question No: 16 (Marks: 1) - Please choose one
Project ---------- defines the concept and range of the proposed solution, and limitations identify certain capabilities that the product will not include

- ► **Scope (Page 30)**
- ► Agreement
- ► Plan
- ► None of the given

Question No: 17 (Marks: 1) - Please choose one
In the N-Tire Architecture the idea is to enhance scalability and ---------- by distributing both the data and the application using multiple server machines.

- ► **Performance (Page 131)**
- ► Efficiency
- ► Usability
- ► non of the all

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Question No: 18 (Marks: 1)

The cyclomatic complexity metric provides the designer with information regarding the number of

- ► Cycles in the program
- ► Errors in the program
- ► **Independent logic paths in the program** **Click here for detail**
- ► Statements in the program

Question No: 19 (Marks: 1)

Which of the followings is not a testing type?

- ► Subsystem testing
- ► Alpha testing
- ► Beta Testing
- ► **Gamma Testing (Page 198)**

Question No: 20 (Marks: 1)

Which of the items listed below is not one of the software engineering layers?

- ► Process
- ► **Manufacturing** **Click here for detail**
- ► Methods
- ► Tools

Question No: 21 (Marks: 1)

Which piece of code is self-documented?

- ►if (x==0) // this is the case when we are allocating a new number
- ►if (AllocFlag == 0)
- ►**If (AllocFlag == NEW_NUMBER) (Page 148)**
- ►None of the given

FINALTERM EXAMINATION
Spring 2010

Question No: 1 (Marks: 1) - Please choose one

UML (unified modeling language) analysis modeling focuses on the _____ .

- ► behavioral model and environment model.
- ► behavioral model and implementation model.
- ► user model and environmental model
- ► **user model and structural model** **(A PRACTITIONER'S APPROACH Page 604)**

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Question No: 2 (Marks: 1) - Please choose one
What makes requirements elicitation difficult?

- ► bounding scope
- ► understanding user needs
- ► requirements volatility
- ► **all of the above (A PRACTITIONER'S APPROACH Page 285)**

Question No: 3 (Marks: 1) - Please choose one
In the architecture trade-off analysis method the architectural style should be described using the

- ► module view
- ► process view
- ► **all of the given (A PRACTITIONER'S APPROACH Page 403)**
- ► data flow view

Question No: 4 (Marks: 1) - Please choose one
In refining the DFD during transaction mapping it is unnecessary to create a PSPEC since only the CSPEC is relevant to this type of architectural style.

- ► True
- ► **False** click here for detail

Question No: 5 (Marks: 1) - Please choose one
A decision table should be used_____

- ► to document all conditional statements
- ► to guide the development of the project management plan
- ► only when building an expert system
- ► **when a complex set of conditions and actions appears in a component** Click here for detail

Question No: 6 (Marks: 1) - Please choose one
Variable names must be in mixed case starting with upper case.

- ► True
- ► **False (Page 150)**

Question No: 7 (Marks: 1) - Please choose one
Which of these are objectives for software testing?

- ► determine the productivity of programmers
- ► eliminate the need for future program maintenance
- ► eliminate every error prior to release
- ► **uncover software errors (A PRACTITIONER'S APPROACH Page 467)**

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Question No: 8 (Marks: 1) - Please choose one
Comparison testing is typically done to test two competing products as part of customer market analysis prior to product release.

- ► True
- ► **False** **Click here for detail**

Question No: 9 (Marks: 1) - Please choose one
In software quality assurance work there is no difference between software verification and software validation

- ► True
- ► **False** **Click here for detail**

Question No: 10 (Marks: 1) - Please choose one
Top-down integration testing has as it's major advantage(s) that

- ► low level modules never need testing
- ► major decision points are tested early
- ► **no stubs need to be written** **Click here for detail**
- ► none of the given

Question No: 11 (Marks: 1) - Please choose one
Rooko mut jane do depicts one of risks associated with inadequate requriements gathering, the name of that risk is ----------

- ► **Ambiguous requirements (Page 20)**
- ► Creeping Requirements
- ► Minimal Specification
- ► Non of the above

Question No: 12 (Marks: 1) - Please choose one
Interaction Diagrams depict the ---------- Behavior of the system

- ► Static
- ► **Dynamic (Page 106)**
- ► Active
- ► Non of the above

Question No: 13 (Marks: 1) - Please choose one
In Sequence Diagrams events are organized in a----------time line

- ► Vertical
- ► **Horizontal (Page 106)**
- ► Vertical & Horizontal
- ► Non of the above

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Question No: 14 (Marks: 1) - Please choose one
the development view describes the ----------organization of the software in its development environment,

- ► **Static (Page 122)**
- ► Dynamic
- ► Still
- ► Static & Dynamic

Question No: 15 (Marks: 1) - Please choose one
Every view is potentially Hierarchical in this regard which of the view are Hierarchical ?

- ► Functioanl View
- ► Development View
- ► C*oncurrency view*
- ► **All of given (Page 125)**

Question No: 16 (Marks: 1) - Please choose one
object-oriented framework that is a set of cooperative classes that make up reusable ---------- of a system

- ► Code
- ► **Design (Page 138)**
- ► Analysis
- ► non of the all

Question No: 17 (Marks: 1) - Please choose one
When planning for performance, one should always remember the ---------- rule

- ► **80/20 (Page 177)**
- ► 20/80
- ► 80/22
- ► 80/90

Question No: 18 (Marks: 1) - Please choose one
In order to make a code more portable, Instead of using vendor specific language extensions, use ---------- as much as possible

- ► **STL (Page 179)**
- ► ANSI
- ► ISO
- ► CMMI

Question No: 19 (Marks: 1) - Please choose one
Performance & Usability are examples of ---------- requirements

- ► Business

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- ► Functioanl
- ► **Non-Functioanl (Page 39)**
- ► User

Question No: 20 (Marks: 1) - Please choose one
Static analyzers are software tools for ---------- processing.

- ► analysis text
- ► **source text (Page 211)**
- ► design text
- ► non of the given

Question No: 21 (Marks: 1) - Please choose one
Software Bugs have multiple names, Which one of the belw is not the name of Software Bugs ----------

- ► Bugs
- ► Defects
- ► Errors
- ► **Mistakes (Page 213)**

Question No: 22 (Marks: 1) - Please choose one
According to a survey, when a software application is in the maintenance phase, --------- of its lifecycle cost is attributed towards the defects which are found in the software application after installation

- ► **20 % (Page 214)**
- ► 25 %
- ► 30%
- ► 35%

Question No: 23 (Marks: 1) - Please choose one
Bugs that won't "----------" (almost random) are the most difficult to deal with.

- ► **stand still (Page 226)**
- ► dynamic
- ► running
- ► stand tall

Question No: 24 (Marks: 1) - Please choose one
1. Variables should be initialized where they are declared and they should be declared in the ---------- scope possible.

- ► **Smallest (Page 158)**
- ► largest
- ► medium
- ► None of the given

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Question No: 25 (Marks: 1) - Please choose one
If you are initializing a variable at the time of declaration, do not declare another ---------- in the same statement.

- ► **variable (Page 176)**
- ► object
- ► class
- ► None of given

Question No: 26 (Marks: 1) - Please choose one
During Requirement Engineering process which of the following is/are document(s) used for fact finding?

- ► Company's employee list
- ► Samples of the company databases
- ► The company's mission statement and plan
- ► **All of given**

FINALTERM EXAMINATION
Fall 2008

Question No: 1 (Marks: 1) - Please choose one
Object-oriented domain analysis is concerned with the identification and specification of reusable capabilities within an application domain.

- ➢ **True** **Click here for detail**
- ➢ False

Question No: 2 (Marks: 1) - Please choose one
Fourth generation techniques

- ➢ Allow software to be developed without any testing.
- ➢ Eliminate the need for costly requirements gathering activities.
- ➢ **Can reduce the time required to develop software.**

(A PRACTITIONER'S APPROACH Page 73)

- ➢ Are best used by non-programmers to build small systems.

Question No: 3 (Marks: 1) - Please choose one
Data design actually begins during the creation of the analysis model, not the architectural model.

- ➢ **True** **(A PRACTITIONER'S APPROACH Page 397)**
- ➢ False

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Question No: 4 (Marks: 1) - Please choose one
The states shown in a state transition diagram do not necessarily correspond to the processes shown in a control flow diagram for the same system.

- **True (A PRACTITIONER'S APPROACH Page 397)**
- False

Question No: 5 (Marks: 1) - Please choose one
The criteria used to assess the quality of an architectural design should be based on system

- **accessibility and reliability (A PRACTITIONER'S APPROACH Page 404)**
- data and control
- functionality
- implementation details

Question No: 6 (Marks: 1) - Please choose one
A useful technique for evaluating the overall complexity of a proposed architecture is to look at the component

- number and size of components
- **flow dependencies and sharing dependencies (A PRACTITIONER'S APPROACH Page 406)**
- size and cost
- none of the given

Question No: 7 (Marks: 1) - Please choose one
The three basic principles that guide maintainability are: simplicity, clarity, and_____ .

- **Generality (Page 146)**
- Reliability
- All of the given choices
- None of the Given choices

Question No: 8 (Marks: 1) - Please choose one
In order to make a program self documented a number of attributes required. Which one is the attribute/s of self documented program

- **All of the given choices (Page 147)**
- Size of each function
- Choice of variable
- Modularity

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Question No: 9 (Marks: 1) - Please choose one
Floating point constants should always be written with decimal point and at least

- **one decimal (Page 161)**
- two decimal
- three decimal
- none of the given

Question No: 10 (Marks: 1) - Please choose one
The code becomes self explanatory with the help of proper use of parentheses. Select the right one.

- leapYear = year % 4 == 0 && year % 100 != 0 || year % 400 == 0 ;
- leapYear = ((year % 4 == 0) && (year % 100 != 0) || ((year % 400 == 0));
- **leapYear = ((year % 4 == 0) && (year % 100 != 0)) || (year % 400 == 0); (Page 164)**
- leapYear = (year % 4 == 0) && (year % 100 != 0) || (year % 400 == 0);
- none of the given options

Question No: 11 (Marks: 1) - Please choose one
Graph-based testing methods can only be used for object-oriented systems

- True
- **False** **Click here for detail**

Question No: 12 (Marks: 1) - Please choose one
Configuration reviews are not needed if regression testing has been rigorously applied during software integration.

- True
- **False** **Click he re for Detail**

Question No: 13 (Marks: 1) - Please choose one
A class is a ---------- of objects

- **Template (Page 85)**
- Stereotype
- Collection
- Non of the above

Question No: 14 (Marks: 1) - Please choose one
A public Interface provides a way for with other Classes.

- **Communication**
- Accessibility
- Reaching
- All of the above

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Question No: 15 (Marks: 1) - Please choose one
Flow Charts represents

- **sequence Activity (Page 50)**
- random activity
- parallel activity
- non of the above

Question No: 16 (Marks: 1) - Please choose one
Software architecture defines the high level structure of the software by putting together a number of architectural ---------- in an organized fashion.

- **Elements (Page 122)**
- parts
- components
- non of the all

Question No: 17 (Marks: 1) - Please choose one
Patterns are devices that allow programs to share knowledge about their ----------.

- **Design (Page 137)**
- Code
- Analysis
- Non of the all

Question No: 18 (Marks: 1) - Please choose one
MVC stands for ----------

- **Model View Controller (Page 140)**
- Modern View Center
- Model View Center
- Modern View Controller

Question No: 19 (Marks: 1) - Please choose one
A ---------- is a code that explains itself without the need of comments and extraneous documentation

- **Self documenting code (Page 147)**
- Self telling Code
- Self Documenting Design
- Non of the them

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Question No: 20 (Marks: 1) - Please choose one
Type conversions must always be done ----------

- **Explicitly (Page 158)**
- implicitly
- simultaneously
- non of them

Question No: 21 (Marks: 1) - Please choose one
Types that are ----------to one file only can be declared inside that file.

- **Local (Page 158)**
- Global
- Private
- General

Question No: 22 (Marks: 1) - Please choose one
The use of do while loops should be ----------

- **Avoided (Page 159)**
- encouraged
- practiced
- non of them

Question No: 23 (Marks: 1) - Please choose one
One of the causes of the portability issues is the order of ---------- varies from one implementation to other.

- **Evaluation (Page 181)**
- numbers
- variables
- symbols

Question No: 24 (Marks: 1) - Please choose one
---------- is a powerful technique that separates error-handling code from normal code.

- **Exception handling (Page 184)**
- Code handling
- variable handling
- pointer handling

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Question No: 25 (Marks: 1) - Please choose one
The goal of testing is to expose ---------- defects in a software system before it is put to use.

- **Latent (Page 193)**
- already present
- current
- runtime

Question No: 26 (Marks: 1) - Please choose one
Static analyzers are software tools for ---------- processing.

- analysis text
- **source text (Page 211)**
- design text
- non of the given

Question No: 27 (Marks: 1) - Please choose one
The first bug was actually a moth, which flew through an open window and into one of the Mark --------- 's relays.

- **II (Page 213)**
- I
- III
- IV

Question No: 28 (Marks: 1) - Please choose one
Which one of the given below is not a symptom of memory overrun?

- Program crashes quite regularly after a given routine is called, that routine should be examined for a possible overrun condition.
- If the routine in question does not appear to have any such problem the most likely cause is that another routine, called in the prior sequence, has already trashed variables or memory blocks.
- Checking the trace log of the called routines leading up to one with the problem will often show up the error.
- **Compiler warnings. (Page 220)**

Question No: 29 (Marks: 1) - Please choose one
Holistic medicine, concerns itself with the state of the body as a whole, not the ---------- that is currently attacking it.

- reason
- **disease (Page 224)**
- source
- non of the given

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Question No: 30 (Marks: 1) - Please choose one
Bugs that won't "stand still" (almost random) are the ---------- to deal with.

- least difficult
- **most difficult (Page 226)**
- very easy
- easy

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